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Crack network evolution in finite domains

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Abstract

The Variational Crack Element Method (VCEM) (Ghoniem 2026a) is a constrained energy-minimization framework that represents cracks through distributed Burgers-density fields, preserving the exact crack-tip singular structure while enabling scalable computation for interacting, branching, and kinked geometries. We establish the numerical fidelity of VCEM through systematic comparison with analytical solutions: straight cracks under mode-I, mode-II, and mixed-mode biaxial loading; circular-arc cracks; and kinked and branched configurations for which complex-variable solutions are available. Stress intensity factors, crack-tip fields, and displacement discontinuities are reproduced with high accuracy across all benchmarks. The method is then applied to the diametrically compressed (Brazilian) disk, a finite-domain problem with strongly non-uniform stress fields. Centerline cracks under force control reproduce the classical splitting–arrest sequence as the tips enter the compressive contact zones; inclined cracks reorient toward the maximum-opening direction with kink angles in quantitative agreement with experiment. Finally, simulations of evolving multi-crack networks demonstrate that fracture under force control can be interpreted as a connectivity phase transition, in which the applied load is the control parameter and the crack-network connectivity is the order parameter, driving the disk from an intact to a fragmented state through unstable growth, junction formation, and branching.

Keywords Variational, Fracture, Crack networks, Crack propagation

Introduction

Fracture is a critical failure mechanism in engineered and natural systems, governing the integrity of structures ranging from aerospace components to geological formations (Anderson 2005; Freund 1990). While classical fracture mechanics provides reliable predictions for single-crack propagation, the modeling of interacting crack networks remains a major challenge due to the nonlocal and collective nature of crack interactions.

From a physics perspective, fracture may be interpreted as a transition driven by the release of stored elastic energy (Griffith 1921; Eshelby 1956; Maugin 1992). Two regimes can be distinguished. In the first, failure is dominated by a single crack front governed by stress intensity factors, as described by linear elastic fracture mechanics (LEFM) (Anderson 2005; Irwin 1957). In the second, failure involves the nucleation and interaction of multiple cracks, leading to distributed damage and eventual coalescence into

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macroscopic fracture. This regime, common in quasi-brittle materials and complex loading environments, cannot be adequately described by single-crack models (Bažant and Cedolin 1991; Herrmann and Roux 1990; Stauffer and Aharony 1994; Sahimi 2003).

This work advances a computational framework for modeling the growth and interaction of crack networks. Cracks are represented as distributed Burgers-density fields, which preserve both the near-tip singular structure and long-range elastic interactions. The formulation is based on a constrained variational principle, in which equilibrium configurations are obtained by minimizing the elastic energy, subject to compatibility, closure, and non-penetration constraints. This framework enables the simulation of kinking, branching, and multi-crack interaction within a unified and computationally tractable approach. The formulation and theoretical foundations are presented in our companion paper (Ghoniem 2026a).

The objective of the present paper is to assess the numerical performance and physical fidelity of the method through systematic validation and application. Benchmark problems include straight, curved, and branched cracks under various loading conditions, for which analytical solutions are available. Following validation, the method is applied to crack evolution in a diametrically compressed disk (Brazilian disk test), a finite-domain problem characterized by strongly inhomogeneous stress fields. This setting provides a controlled framework to examine the influence of loading conditions on crack stability and propagation and to interpret fracture behavior in terms of evolving energy landscapes.

The remainder of the paper is organized as follows. [Method validation](#) section presents validation against analytical solutions for straight, mixed-mode, branched, and curved cracks. [Parametrization of crack networks](#) section examines the robustness of different crack parameterizations. [Simulations of disk compression experiments](#) section applies the method to the Brazilian disk problem, including comparisons with experimental observations. [Fracture as a “connectivity phase transition”](#) section investigates the evolution of complex crack networks and fragmentation behavior. Concluding remarks are given in [Conclusions](#) section.

Summary of the VCEM formulation

We summarize here the Variational Crack Element Method (VCEM) developed in the companion formulation paper (Ghoniem 2026a), to which the reader is referred for derivations and implementation details. VCEM treats interacting crack networks in two-dimensional isotropic linear elasticity as a constrained energy-minimization problem at fixed geometry, with crack extension handled separately through local near-tip criteria.

Crack-network representation

The network is encoded as an undirected graph $\mathcal{G} = (V, E)$ with a geometric embedding $\Gamma = \bigcup_{e \in E} \Gamma_e$. Vertex degree classifies the topology: $\deg(v) = 1$ are crack tips, $\deg(v) = 2$ are kinks, and $\deg(v) \geq 3$ are junctions. Each edge carries a local frame $(e_1^{(e)}, e_2^{(e)})$ and a polarity $\pi^{(e)} \in \{+1, -1\}$ assigned by the right-hand rule.

Crack-face kinematics are represented by the displacement jump $J(s) = u^+ - u^-$, parameterized by a distributed Burgers density

$$\frac{dJ}{ds}(s) = b_{II}(s) t(s) + b_I(s) n(s), \quad J(s) = \int_0^s [b_{II}t + b_I n] d\tilde{s}, \quad (1)$$

where b_I and b_{II} generate Mode I (opening, $COD = J \cdot n$) and Mode II (sliding, $CSD = J \cdot t$) discontinuities, respectively. The *Burgers content vector* $B = \int_{\Gamma} (b_{II}t + b_I n) ds = J(L) - J(0)$ distinguishes dipolar cracks ($B = 0$) from polar ones ($B \neq 0$). Three element representations are supported: piecewise-constant uniform, Chebyshev–Lobatto clustered, and singular (with embedded $1/\sqrt{L-s}$ or $1/\sqrt{s(L-s)}$ tip asymptotics).

Elastic kernels

The displacement and stress fields are derived from the Kelvin fundamental solution U_{ij} through its spatial derivatives. With $\Delta = x - \xi$, $R = \|\Delta\|$, and $M = \mu/[2\pi(1-\nu)]$ (plane strain), the elastostatic dislocation kernel takes the closed form

$$K_{ij\alpha}(x, \xi) = M \left[\frac{\delta_{i\alpha}\Delta_j + \delta_{j\alpha}\Delta_i - \delta_{ij}\Delta_\alpha}{R^2} - \frac{2\Delta_i\Delta_j\Delta_\alpha}{R^4} \right], \quad (2)$$

so that the stress field generated by the distributed density is $\sigma_{ij}(x) = \sigma_{ij}^\infty(x) + \int_{\Gamma} K_{ij\alpha}(x, \xi) b_\alpha(\xi) d\Gamma_\xi$.

Constrained variational equilibrium

For a fixed geometry, the Burgers density minimizes the elastic potential energy

$$\Phi[b] = \frac{1}{2} \int_{\Gamma} \int_{\Gamma} b_\beta(x) T_{\beta\alpha}(x, \xi) b_\alpha(\xi) d\Gamma_\xi d\Gamma_x - \int_{\Gamma} b_\beta(x) t_\beta^{\text{app}}(x) d\Gamma_x, \quad (3)$$

subject to admissibility constraints that encode the topology of $\mathcal{G}$: (i) *junction closure*, $\sum_{b=1}^m B^{(b)}(x_0) = 0$ at every vertex with $m \geq 3$, ensuring single-valued displacement and Burgers-vector conservation; (ii) *Burgers-vector/gauge conditions* at tips and prescribed endpoints; and (iii) *unilateral non-penetration*, $COD(s) \geq 0$.

Discretizing $b(s) \approx \sum_i q_i N_i(s)$ and assembling the equality constraints into $C_J q = 0$ and the active normal-contact constraints into $C_N^{(A)} q = 0$ yields the Karush–Kuhn–Tucker (KKT) saddle-point system

$$\begin{bmatrix} H & C_J^T & C_N^{(A)T} \\ C_J & 0 & 0 \\ C_N^{(A)} & 0 & 0 \end{bmatrix} \begin{Bmatrix} q \\ \lambda \\ \mu \end{Bmatrix} = \begin{Bmatrix} f \\ 0 \\ 0 \end{Bmatrix}, \quad \mu \geq 0, \quad \mu^T (C_N q) = 0. \quad (4)$$

Two equivalent realizations are available for H and f : a Galerkin (energy-based) form with $H = A$ symmetric positive semi-definite, and a least-squares (collocation) form with $H = K^T K + \rho I$ and $f = K^T f_{\text{coll}}$. Junction compatibility can be enforced in *strict* mode (exact closure via C_J), *core* mode (restriction to $q \in \mathcal{N}(C_J)$), or *soft* mode (penalty $\frac{\eta}{2} \|C_J q\|^2$). Strict enforcement eliminates artificial incompatibility singularities at junctions, so that $r^{-1/2}$ behavior is confined to physical $\deg(v) = 1$ tips.

Finite-domain coupling

Bounded domains are handled by coupling VCEM to a boundary element (BEM) representation of the outer boundary based on the same Kelvin kernels. Crack-induced traces on Γ are mapped through operators M_u, M_t , and the resulting augmented system

$$\begin{bmatrix} K_{qq} & N & C_{vc}^T & M^T \\ 0 & C_{bem} & 0 & C_{bc}^T \\ C_{vc} & 0 & 0 & 0 \\ M & C_{bc} & 0 & 0 \end{bmatrix} \begin{Bmatrix} q \\ y \\ \lambda \\ \mu \end{Bmatrix} = \begin{Bmatrix} f_{vc} \\ 0 \\ g_{vc} \\ d \end{Bmatrix} \quad (5)$$

solves simultaneously for crack DOFs q and boundary-correction DOFs y . Three operational modes are supported: uncoupled (one-way BEM $\rightarrow$ VCEM), iterative coupling, and direct monolithic solution of (5).

Crack-network evolution

Once equilibrium is obtained, each tip is advanced by a small increment Δa using near-tip quantities extracted from the converged fields. Stress-intensity factors are computed by a two-term Williams fit (Williams 1956) to the near-tip COD/CSD, $K_I = \mu A_I/(\kappa + 1)$, $K_{II} = \mu A_{II}/(\kappa + 1)$, or from the configurational Peach–Koehler force projected onto the local tip frame, with the inversion $J_1 = (K_I^2 + K_{II}^2)/E'$, $J_2 = -2K_I K_{II}/E'$. The propagation angle θ follows from a standard mixed-mode criterion (MTS, SED, or local symmetry); for MTS,

$$\theta_{MTS} = 2 \arctan \left[\frac{1}{4} \left(\frac{K_I}{K_{II}} \pm \sqrt{(K_I/K_{II})^2 + 8} \right) \right]. \quad (6)$$

Propagation occurs when $K_{eff} = \sqrt{K_I^2 + K_{II}^2} \geq K_c$ (deterministic) or when $K_{eff}(x_{tip}) > K_c(x_{tip})$ with K_c sampled from a prescribed random field (stochastic, microstructure-aware). After each step, the polyline is extended by a polar half-crack element at the active tip, the graph is updated, and the equilibrium problem (4) (or (5)) is re-solved.

This constrained variational structure — Burgers-density crack kinematics (1), Kelvin-based elastic kernels (2), a KKT saddle-point equilibrium (4) with strict junction closure and unilateral contact, optional BEM coupling (5), and criterion-based extension (6) — forms the computational backbone used in the applications developed in the remainder of this paper.

A schematic overview of the method (Ghoniem 2026a) is shown in Fig. 1.

Stress intensity factors

Before presenting the convergence results, we describe how stress intensity factors (SIFs) are extracted from the computed Burgers density. Three complementary estimators are implemented: a near-tip fit to the displacement jump (K_{jump}), a near-tip fit to the stress field (K_{tip}), and a configurational Peach–Koehler force evaluation (K_{pk}). All three derive from the same Burgers density $b(s)$ obtained from the equilibrium solve of Eq. (4); they differ in whether they probe the primary displacement jump, the derived stress field, or the associated energy release rate. Their joint use provides an internal consistency check on the accuracy of the near-tip solution.

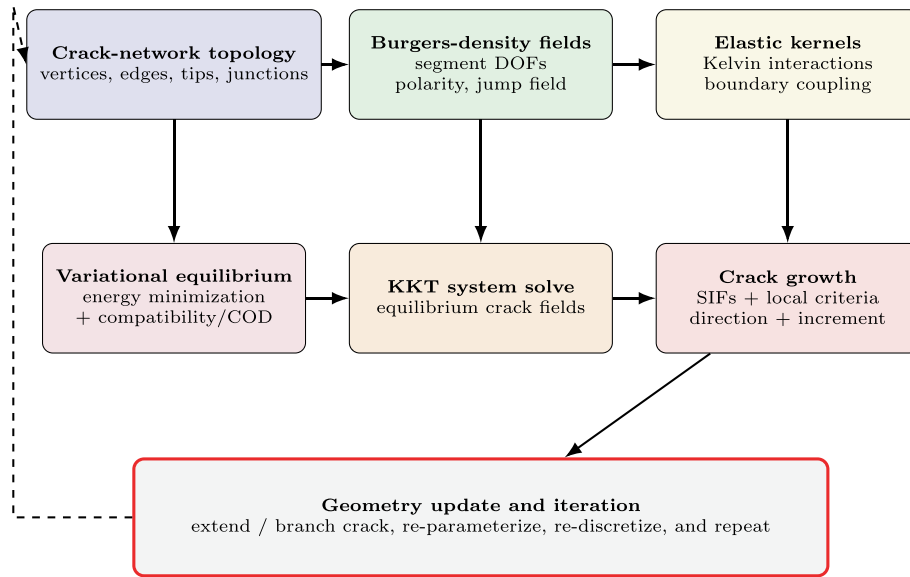


Fig. 1 Schematic of the Variational Crack Element Method (VCEM), emphasizing the two-layer structure (representation and solution) and iterative crack-network evolution

(a) Displacement-jump estimator K_{jump} . In VCEM, the displacement jump across a crack is not an independent unknown but the path integral of the Burgers density (Eq. (1)),

$$J(s) = \int_0^s [b_{II}(\tilde{s}) t(\tilde{s}) + b_I(\tilde{s}) n(\tilde{s})] d\tilde{s}, \quad \text{COD}(s) = J(s) \cdot n(s), \quad \text{CSD}(s) = J(s) \cdot t(s). \quad (7)$$

The Williams expansion (Williams 1956) of the near-tip displacement field gives the crack-face discontinuities as

$$\text{COD}(r) = A_I \phi(r) + B_I \phi(r) \bar{r} + \mathcal{O}(r^{3/2}), \quad \text{CSD}(r) = A_{II} \phi(r) + B_{II} \phi(r) \bar{r} + \mathcal{O}(r^{3/2}), \quad (8)$$

where r is the distance from the active tip measured along the crack face, $\phi(r) = \sqrt{r/(2\Phi)}$, $\bar{r} = r/(2\Phi)$, and Φ is the local tip half-length scale used for non-dimensionalisation. The linear-in- r terms absorb the subleading regular contribution and improve the stability of the fit over a finite window. Because $\text{COD}(r)$ and $\text{CSD}(r)$ are known analytically from the computed Burgers density through Eq. (7), the amplitudes A_I , A_{II} are obtained by least-squares regression of the numerical jump profiles onto the basis $\{\phi(r), \phi(r) \bar{r}\}$ over a tip window of length $r_{\max}$. The SIFs follow from the Irwin/Kolosov relations,

$$\boxed{K_I^{\text{jump}} = \frac{\mu}{\kappa + 1} A_I, \quad K_{II}^{\text{jump}} = \frac{\mu}{\kappa + 1} A_{II},} \quad (9)$$

with μ the shear modulus and $\kappa = 3 - 4\nu$ in plane strain or $(3 - \nu)/(1 + \nu)$ in plane stress. The singular element representation of Sec. 3.1 of our paper (Ghoniem 2026a) embeds the $1/\sqrt{L-s}$ (half-crack) or $1/\sqrt{s(L-s)}$ (full-crack) tip asymptotics directly into the basis functions, so that $A_{I,II}$ is recovered with controlled error as $r_{\max}$ is refined.

(b) Near-tip stress estimator K_{tip} . The companion Williams expansion of the stress field on the crack-line continuation ($\theta = 0$, $r > 0$ ahead of the tip) reads

$$\sigma_{22}(r, 0) = \frac{K_I}{\sqrt{2\pi r}} + \mathcal{O}(1), \quad \sigma_{12}(r, 0) = \frac{K_{II}}{\sqrt{2\pi r}} + \mathcal{O}(1). \quad (10)$$

In VCEM, the stress components $\sigma_{22}(r, 0)$ and $\sigma_{12}(r, 0)$ are evaluated from the computed Burgers density through the Kelvin-based dislocation kernel (Eq. (2)) at a set of sample points $r_q \in (\varepsilon, r_{\max}]$ arranged ahead of the active tip. The inner cutoff ε is chosen a few quadrature-subinterval widths away from the tip so as to exclude the self-panel contribution and avoid the logarithmic/principal-value singularity of the kernel at $r \rightarrow 0$. Regressing the computed stresses on $1/\sqrt{r}$,

$$\sigma_{22}(r_q, 0) \approx \frac{C_I}{\sqrt{r_q}} + D_I, \quad \sigma_{12}(r_q, 0) \approx \frac{C_{II}}{\sqrt{r_q}} + D_{II}, \quad (11)$$

the singular amplitudes $C_{I,II}$ are extracted by least squares and the SIFs follow directly from Eq. (10):

$$\boxed{K_I^{\text{tip}} = \sqrt{2\pi} C_I, \quad K_{II}^{\text{tip}} = \sqrt{2\pi} C_{II}.} \quad (12)$$

The K_{tip} estimator is the stress-based counterpart of K_{jump} : both rely on the Williams expansion, but K_{jump} fits the square-root part of the *primary* field (the displacement discontinuity, which is itself $\mathcal{O}(\sqrt{r})$ and therefore well resolved), whereas K_{tip} fits the square-root *singular* part of a derived field and is consequently more sensitive to the windowing parameters $(\varepsilon, r_{\max})$ and to near-tip discretisation.

(c) Peach–Koehler estimator K_{pk} . The third estimator is energetic rather than asymptotic. Treating the distributed Burgers density as a continuous array of Volterra dislocations, the configurational (Peach–Koehler) force on the crack tip is accumulated over a near-tip panel window $\mathcal{W}$:

$$\mathbf{F}_{PK} \approx \sum_{k \in \mathcal{W}} (\boldsymbol{\sigma}(\mathbf{x}_k) d\mathbf{B}_k) \times \mathbf{e}_z, \quad d\mathbf{B}_k = (b_{II,k} \mathbf{t}_k + b_{I,k} \mathbf{n}_k) \Delta s_k, \quad (13)$$

where $\boldsymbol{\sigma}(\mathbf{x}_k)$ is the total stress at the k -th PK integration point, evaluated from Eq. (2) with the self-contribution of the window itself removed so as to exclude the singular self-stress. Projecting $\mathbf{F}_{PK}$ onto the local tip frame yields the components of the J -integral,

$$J_1 = \mathbf{F}_{PK} \cdot \mathbf{t}, \quad J_2 = \mathbf{F}_{PK} \cdot \mathbf{n}, \quad (14)$$

which in two-dimensional linear elastic fracture mechanics are related to the SIFs by the classical Irwin/Rice relations

$$J_1 = \frac{K_I^2 + K_{II}^2}{E'}, \quad J_2 = -\frac{2 K_I K_{II}}{E'}, \quad E' = \begin{cases} E & \text{(plane stress),} \\ E/(1 - \nu^2) & \text{(plane strain).} \end{cases} \quad (15)$$

Introducing $A = E' J_1$, $C = -\frac{1}{2} E' J_2$, and $D = \sqrt{A^2 - 4C^2}$, the system (15) is inverted in closed form to give

$$\boxed{K_I^{\text{pk}} = \sqrt{\frac{1}{2}(A + D)}, \quad K_{II}^{\text{pk}} = \text{sign}(C) \sqrt{\frac{1}{2}(A - D)}.} \quad (16)$$

The sign of K_{II}^{pk} is fixed by the sign of J_2 , while $J_1 \geq 0$ is automatic since $F_{PK} \cdot t$ is the physical energy release rate associated with tangential tip advance. For curved cracks (e.g. the circular-arc case of [Circular arc crack](#) section) the SIFs obtained from the local tip frame are rotated to the global frame through the Cotterell–Rice kink-angle relations, reproduced as Eq. (102) of our paper ([Ghoniem 2026a](#)).

Consistency among the three estimators. Equations (9), (12), and (16) are mathematically equivalent in the limit of infinite near-tip refinement: the first two are the two Williams readouts of the same asymptotic field (the first from the primary displacement discontinuity, the second from the derived stress), while the third recovers the same SIFs through the energy release rate via (15). The convergence of all three to a common limit in the benchmark problems of [Method validation](#) section is therefore a non-trivial internal consistency check: K_{jump} probes the solution for the *primary* unknown $b(s)$, whereas K_{tip} and K_{pk} probe two different *derived* quantities of the same solution. Simultaneous agreement among the three confirms that both the Burgers density and the Kelvin-based stress reconstruction are correctly resolved at the tip.

Iterative BEM–KKT solver: implementation procedure

The compressed-disk simulations use the iterative mode of the BEM–KKT coupling introduced in Sec. 4.6 of our companion paper ([Ghoniem 2026a](#)), in which the crack solve and the outer-boundary solve are alternated within each crack-growth cycle until the coupled system converges. This iterative scheme avoids assembling and factorizing the full monolithic system of Eq. 5 at every cycle while still capturing the two-way coupling between the crack-induced fields and the bounded elastic domain. The implementation proceeds as follows.

Step 1 — Pre-cycle assembly of the BEM operators. For the current outer-boundary discretization ($N_{bem} = 120$ panels in the disk simulations), the dense single- and double-layer matrices G and H of Eqs. (84)–(85) of our paper ([Ghoniem 2026a](#)) are assembled once at the start of each crack-growth cycle, using $N_g^{bem} = 6$ Gauss points per panel and a Cauchy principal-value treatment of the diagonal H entries. Imposing the prescribed boundary conditions on the disk (radial pressure on the loading arcs, traction-free elsewhere) and performing the standard unknown-swapping yields the BEM block C_{bem} of Eq. (86) and the right-hand side r_{bem} for the uncracked configuration. This factorization is reused throughout the inner iteration of the current cycle.

Step 2 — Crack-to-boundary coupling operators. The crack-to-boundary trace operators M_u and M_t of Eq. (89) of our paper ([Ghoniem 2026a](#)) are assembled by evaluating the displacement and traction of every crack basis function at every BEM collocation point through the Kelvin kernels of Eq. (2). Both are dense matrices of size $N_{bem} \times N_{dof}^{cr}$, where N_{dof}^{cr} is the number of crack DOFs at the current cycle. They are reassembled at the start of every cycle because the crack geometry has changed but are held fixed during the inner iteration.

Step 3 — Initialization. The inner iteration is initialized with the crack-induced traces set to zero, $u_{\Gamma}^{\text{cr},(0)} = 0$ and $t_{\Gamma}^{\text{cr},(0)} = 0$, so that the first sub-iterate corresponds to the standalone BEM solve for the uncracked disk under the applied loading.

Step 4 — BEM sub-solve at iteration k . Given the current crack-induced traces, the BEM unknowns $y^{(k)} = [u^{bc,(k)}; t^{bc,(k)}]$ are obtained from

$$C_{\text{bem}} y^{(k)} = r_{\text{bem}} - H u_{\Gamma}^{\text{cr},(k-1)} + G t_{\Gamma}^{\text{cr},(k-1)}, \quad (17)$$

which is the discrete form of the boundary integral identity (86) augmented by the crack-induced contribution. Equation (17) is solved by direct LU factorisation reused from Step 1.

Step 5 — Recovery of the boundary-induced traction on the crack. The boundary correction $y^{(k)}$ is mapped back onto the crack as the additional applied traction

$$t^{\text{ext},(k)} = t^{\text{app}} + N y^{(k)}, \quad (18)$$

where N is the boundary-to-crack traction map (Sec. 4.6 of our paper (Ghoniem 2026a), Eq. (93)). The first term t^{app} is the infinite-medium applied traction (which vanishes for the disk problem since loading is applied only on the boundary); the second term encodes the feedback of the bounded domain on the crack network.

Step 6 — Crack KKT sub-solve at iteration k . With the updated right-hand side $f^{(k)}$ assembled from $t^{\text{ext},(k)}$, the constrained variational problem (4) is solved on the current crack network for the Burgers density $q^{(k)}$ and the associated Lagrange multipliers,

$$\begin{bmatrix} H_{cr} & C_J^T & C_N^{(\mathcal{A})T} \\ C_J & 0 & 0 \\ C_N^{(\mathcal{A})} & 0 & 0 \end{bmatrix} \begin{Bmatrix} q^{(k)} \\ \lambda^{(k)} \\ \mu^{(k)} \end{Bmatrix} = \begin{Bmatrix} f^{(k)} \\ 0 \\ 0 \end{Bmatrix}, \quad \mu^{(k)} \geq 0. \quad (19)$$

The active-set strategy of Sec. 4.5 of our paper (Ghoniem 2026a) is applied to the unilateral non-penetration constraints $C_N q \geq 0$ within this sub-solve: starting from the active set of the previous iteration, the equality-constrained KKT system is solved, the multipliers $\mu^{(k)}$ are checked for sign, and constraints are added to or removed from the active set until the complementarity conditions are satisfied. Convergence of the active set is typically achieved in 1–3 inner sweeps.

Step 7 — Update of the crack-induced traces. Once $q^{(k)}$ has been computed, the crack-induced traces on the outer boundary are updated using the coupling operators of Step 2,

$$u_{\Gamma}^{\text{cr},(k)} = M_u q^{(k)}, \quad t_{\Gamma}^{\text{cr},(k)} = M_t q^{(k)}. \quad (20)$$

These are then fed back into Step 4 for the next BEM sub-solve.

Step 8 — Convergence test. The inner iteration is terminated when both the crack-DOF and the BEM correction have converged in a relative L^2 sense:

$$\frac{\|q^{(k)} - q^{(k-1)}\|_2}{\|q^{(k)}\|_2} < \varepsilon_q, \quad \frac{\|y^{(k)} - y^{(k-1)}\|_2}{\|y^{(k)}\|_2} < \varepsilon_y, \quad (21)$$

with $\varepsilon_q = \varepsilon_y = 10^{-6}$ in the simulations of [Centerline slit: growth, closure, and shear localization](#) and [Compressed disk with an off-axis slit](#) sections. In all reported cases, convergence was achieved in 3–6 inner iterations per crack-growth cycle, with the crack geometry fixed during the inner iteration and updated only once the inner loop has converged. After convergence, the SIFs are extracted at every active tip using the procedures, and the crack-growth algorithm of [Crack growth algorithm](#) section advances the geometry to the next cycle.

Step 9 — Numerical robustness. Two implementation details are worth recording. First, the BEM matrix C_{bem} and the crack stiffness H_{cr} are scaled by characteristic moduli G and lengths L before the assembly of the coupled right-hand side, following the non-dimensionalization of Sec. 4.6 of our paper (Ghoniem 2026a) in order to bring the two blocks to comparable magnitudes and avoid loss of precision when forming Ny . Second, a mild Tikhonov regularization ρI with $\rho = 10^{-10} \|H_{cr}\|_F$ is added to the crack stiffness in the least-squares form (Sec. 4.4.2 of our paper (Ghoniem 2026a)) to control conditioning when the active set toggles between inner sub-iterations; this regularization is small enough to leave the solution unchanged to graphical accuracy but is sufficient to prevent spurious oscillations of the active set.

Crack mesh options

Figure 2 shows the three discretization strategies compared in this work — uniform (a), tip-dense Chebyshev–Lobatto (b), and the singular-tip representation with embedded $1/\sqrt{r}$ asymptotics (c) — at the coarsest convergence-study resolution $N_{\text{half}}^e = 25$. Panels (d) and (e) show the same tip-dense node layout applied to the inclined-crack benchmark of [Mixed mode straight crack](#) section and the Y-branched-crack benchmark of [Branched & kinked cracks](#) section, with main tip, junction, and branch tip marked.

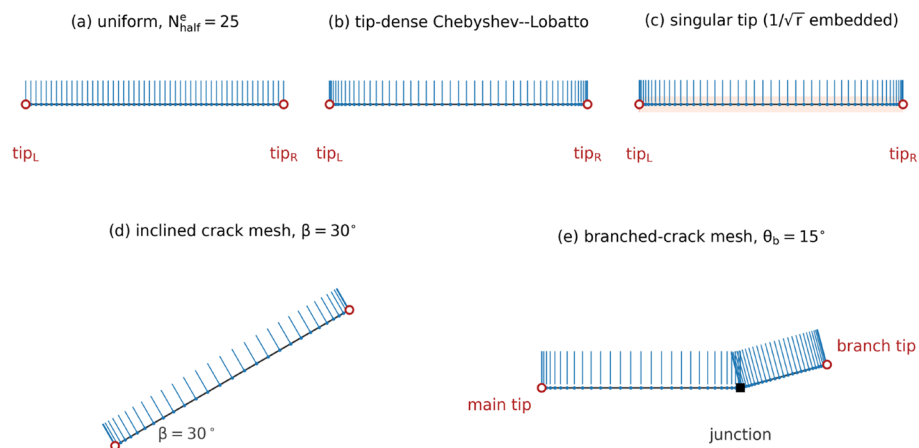


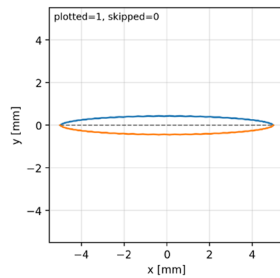
Fig. 2 Representative crack-element discretizations at the coarsest resolution of the convergence study, $N_{\text{half}}^e = 25$. **a** Uniform spacing along the crack curve. **b** Tip-dense Chebyshev–Lobatto clustering, which concentrates collocation nodes near each tip with a quadratic spacing in physical arc length. **c** Singular tip representation used by the VCM: the same tip-dense node distribution as in (b), with the analytic singular weight $1/\sqrt{(s)(L-s)}$ embedded in the kernel quadrature; the red band is a visual aid showing the spatial decay of this weight away from the two tips. **d** Mesh used for the inclined-crack benchmark of [Method validation](#) section, $\beta = 30^\circ$. **e** Mesh used for the Y-branched-crack benchmark of [Method validation](#) section, showing the main crack, the junction node (filled square), and the single branch at $\theta_b = 15^\circ$. The collocation nodes for (d) and (e) follow the tip-dense Chebyshev–Lobatto distribution of (b)

Method validation

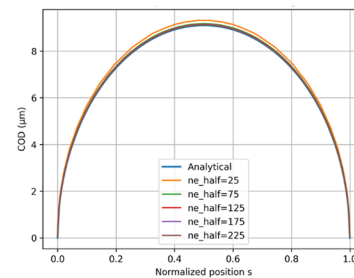
In all validation results in this section, the elastic modulus was taken as $E=200$ GPa and Poisson's ratio of $\nu = 0.30$. Other conditions will be highlighted as necessary. N_{half}^e is the number of elements on one half of the crack, with the convention $N_{\text{element}} = 2N_{\text{half}}^e$.

Single-mode straight crack

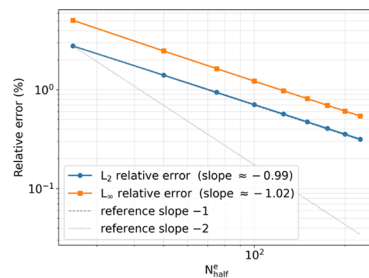
To validate the implementation, we consider a single straight crack under remote mode-I tension and compare the numerical results with classical LEFM solutions. For this benchmark, the reference stress intensity factor is $K_I^{\text{ref}} = \sigma_{\infty} \sqrt{\pi c}$, where c is the crack half-length. The reference crack-opening profile has the expected square-root endpoint behavior and a maximum at the center of the crack. In Fig. 3a ‘plotted=X’ indicate the number of data records actually rendered in the plot and “skipped=Y” the number that were omitted because they fell outside the plot bounds.



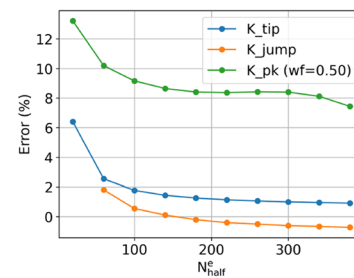
(a) Geometry of the straight-crack example.



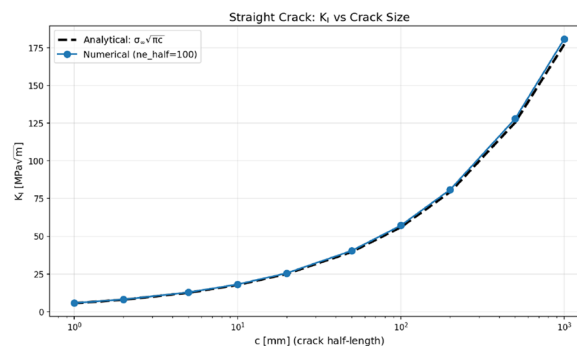
(b) Crack opening displacement (COD) along the crack faces for different numbers of elements, $N_{\text{element}} = 2N_{\text{half}}^e$.



(c) RMS error in COD as a function of N_{half}^e .



(d) Convergence of the stress intensity factor (SIF) evaluated using three methods.



(e) Dependence of the mode-I SIF, K_I , on crack length over several decades.

Fig. 3 Validation plots for the numerical algorithm applied to a straight crack under mode-I loading

Figure 3b shows that the computed COD profiles converge monotonically to the analytical curve as the discretization is refined. The largest mismatch occurs near the tips, where the singular field is strongest, while the agreement in the interior is already good for moderate resolution. Figure 3c confirms the quantitative convergence of the COD error with increasing N_{half}^e : both L_2 and L_∞ errors decay steadily, with L_∞ remaining larger, as expected for a tip-dominated norm. Figure 3d compares the SIF estimates obtained by three post-processing routes (K_{tip} , K_{jump} , and K_{pk}) that correspond to the Williams expansion tip fit method, the Williams expansion tip displacement gradient (jump) method, and the Peach-Koehler force estimation from the local field method, respectively. All methods converge with refinement; the jump-fit estimate is the most accurate in this pure mode-I case, while the tip- and PK-based estimates are slightly more sensitive to near-tip discretization and windowing at coarse meshes.

The quantity plotted in Fig. 3d is the signed point-wise relative error of the tip SIF, $e_K = 100 ([K_{\text{num}} - K_{\text{ref}}]/K_{\text{ref}}) [\%]$, with K_{ref} being the closed-form plane-strain solution. The negative sign of the curve labeled K_{jump} in Fig. 3(d) reflects a known bias of one-point COD-extrapolation estimators (Ingraffea and Manu 1980; Aliabadi 2002). Writing $\delta_{\text{ana}}(r) = (4K/E^*)\sqrt{r/(2\pi)}[1 + c_1(r/c) + \mathcal{O}(r^2/c^2)]$ for a center crack of half-length a under uniform remote tension, with $c_1 = -1/4 < 0$, the jump estimator evaluated at the first interior collocation node $r_1 \propto h_{\text{tip}}$ yields $K_{\text{jump}} = K[1 + c_1(r_1/c) + \mathcal{O}(r_1^2/c^2)]$, so $K_{\text{jump}} < K$ and the relative error is strictly negative for any finite mesh, decaying as $\mathcal{O}(h_{\text{tip}})$. The tip-asymptotic and Peach-Koehler estimators, which act on the full singular field rather than on a single reconstructed value, do not exhibit this bias and converge to zero error monotonically. Finally, Fig. 3e verifies the size-scaling law over several decades of crack-length. The numerical K_I closely follows the analytical trend $\propto \sqrt{c}$, demonstrating that the solver preserves the correct LEFM scaling, not only in a single geometry but across a wide range of crack sizes.

In FEM-based fracture analysis, the $r^{1/2}$ asymptotic behavior of the crack-opening displacement near the tip is not naturally captured by standard polynomial shape functions, and special elements (Barsoum quarter-point isoparametric elements (Barsoum 1976), singular enrichment in XFEM, etc.) are introduced to recover the correct singularity. The VCEM does not require such regularization, for two reasons that are intrinsic to the formulation. First, the unknown solved for is the Burgers density $D(s)$ along the crack curve, and the analytic singular weight $w(s) = 1/\sqrt{s(L-s)}$ for a fully embedded crack (or $1/\sqrt{L-s}$ for an edge crack) is incorporated directly into the kernel quadrature. Consequently the discretized unknown is the regular part $\tilde{D}(s) = w^{-1}(s)D(s)$, which is smooth up to the tip; the $r^{-1/2}$ stress singularity and the $r^{1/2}$ COD behavior are recovered exactly when D is reconstructed. Second, the source nodes are clustered toward each tip with a power law $\xi = u^p$, $p = 2$ by default, providing further geometric refinement where curvature in $\tilde{D}$ is largest. The combination of analytic singular weighting and tip-clustered nodes is in fact stronger than the Barsoum construction (Barsoum 1976), which only approximates $r^{1/2}$ through quarter-point node placement: in the VCEM the singular behavior is exact for any N_{half}^e , and increasing N_{half}^e refines only the smooth correction $\tilde{D}$.

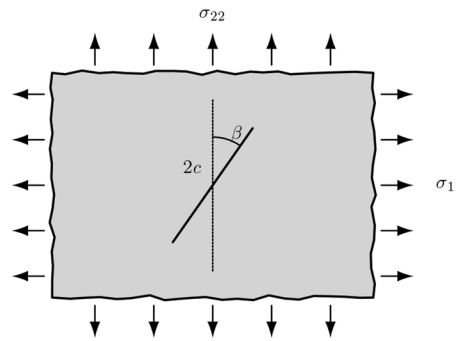
Mixed mode straight crack

Closed-form expressions for the stress intensity factors of an inclined crack of length $2a$ under remote biaxial loading are available (Rooke and Cartwright 1976). For a crack oriented at an angle β with respect to the loading axis,

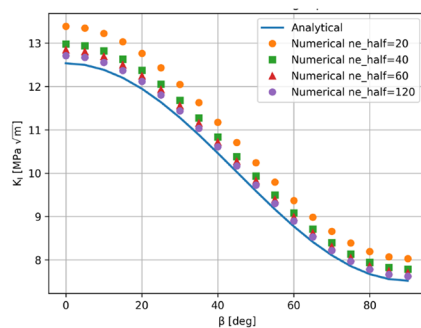
$$K_I = \sqrt{\pi c} (\sigma_{22} \cos^2 \beta + \sigma_{11} \sin^2 \beta), \quad (22)$$

$$K_{II} = \sqrt{\pi c} (\sigma_{22} - \sigma_{11}) \sin \beta \cos \beta. \quad (23)$$

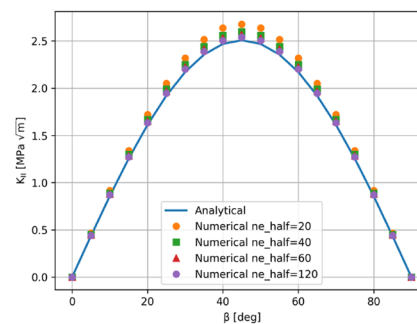
Figure 4a compares the computed mode-I SIF with the analytical prediction over $0^\circ \leq \beta \leq 90^\circ$ for several discretizations ($N_{\text{element}} = 2N_{\text{half}}^e$). The numerical K_I decreases monotonically with increasing β , consistent with the analytical dependence $\cos^2 \beta - \sin^2 \beta$. The convergence toward the closed-form solution is uniform with mesh refinement. Even at moderate resolution, the discrepancy is small, and at the finest discretization, the numerical and analytical curves are nearly indistinguishable over the full angular range. Figure 4b shows the corresponding results for K_{II} . The computed shear intensity reproduces the expected variation $\sin \beta \cos \beta$, vanishing at $\beta = 0^\circ$ and 90° , and reaching a maximum near $\beta = 45^\circ$. The peak magnitude and its angular location are accurately captured. As in the mode-I case, the solution exhibits systematic convergence with increasing N_{half}^e , with residual deviations near the maximum diminishing under refinement.



(a) Geometry and loading configuration for the inclined crack under biaxial loading.



(b) K_I versus the crack inclination angle β for several discretizations ($N_{\text{element}} = 2N_{\text{half}}^e$).



(c) K_{II} versus the crack inclination angle β for several discretizations ($N_{\text{element}} = 2N_{\text{half}}^e$).

Fig. 4 Mixed-mode validation for an inclined crack under biaxial loading. **a** Problem geometry and loading configuration. **b, c** Comparison of numerical stress intensity factors with analytical predictions over $0^\circ \leq \beta \leq 90^\circ$

In general, the mixed-mode benchmark confirms that the present discretization and SIF extraction procedures accurately capture both the angular dependence and the $\sqrt{c}$ scaling of the stress intensity factors under biaxial loading. The numerical method reproduces the expected transition from opening-dominated to shear-dominated response as the crack rotates, providing strong validation for subsequent mixed-mode and networked-crack simulations in finite domains.

Branched & kinked cracks

An essential feature of crack networks is the precise growth direction and magnitude of new segments at deg-1 vertices (crack tips). Although growth criteria can be model dependent if one wants to simulate crack growth in random media or materials with strong microstructure variations, one of the key elements is the accuracy of the stress field at the tips of cracks and, in particular, the accurate determination of the stress intensity factor (SIF). The analytical treatment of branched and curved cracks has evolved through a combination of complex-variable elasticity, singular integral equation methods, perturbation theory, and numerical approaches. The principal contributions of the works considered here are summarized below.

Early foundational work on crack branching was carried out by Congleton and Petch (1967), who examined the conditions under which a propagating crack bifurcates. Their analysis combined fracture mechanics criteria with experimental observations, establishing the importance of local stress fields and branching angle selection in brittle materials. Sih and co-workers introduced energy-density criteria and mixed-mode fracture formulations that later influenced branching stability analysis. These ideas formed the basis for later treatments of bifurcation and crack deflection problems. Lo (1978) provided one of the first systematic closed-form solutions for symmetric branched cracks using complex potential methods. The approach relied on Muskhelishvili's formulation of plane elasticity and conformal mapping to reduce the geometry to a tractable form.

Lo (1978) derived stress intensity factors for symmetric branch configurations and quantified their dependence on the branch-length ratio, providing benchmark solutions for subsequent numerical validation. Kitagawa (1977) extended the analysis to branched cracks under biaxial loading, employing complex potentials and singular integral equation techniques. Their work clarified the interaction between the main crack and secondary branches under mixed-mode loading conditions. Vitek (1977) computed plane-strain stress intensity factors for branched cracks using integral-equation methods. The formulation emphasized accurate resolution of the square-root singularity near each branch tip and provided tabulated SIF values that became widely cited reference data. Isida and Noguchi (1992) later refined the evaluation of stress intensity factors at branched crack tips under general loading. Their method combined complex function theory with numerical evaluation of singular integrals, achieving improved accuracy for arbitrary branch angles and loading combinations. This work is frequently used as a benchmark for finite-element and boundary-element validations. Dollar and Steif (1991) revisited the branched crack problem, emphasizing analytical consistency and comparison with earlier results. They clarified discrepancies between prior solutions and re-examined the limiting behavior as branch length tends to zero. Katzav et al. (2007) addressed dynamic crack branching using linear elastic fracture mechanics combined with stability analysis. Their formulation incorporated dynamic stress intensity factors and energy balance

considerations, establishing conditions for spontaneous branching in rapidly propagating cracks. Rassoulova (2010) investigated star-shaped and multi-branched crack geometries. Using complex potential methods and eigenfunction expansions, she obtained stress intensity factors for multi-tip configurations, extending classical two-branch solutions to higher-order symmetry.

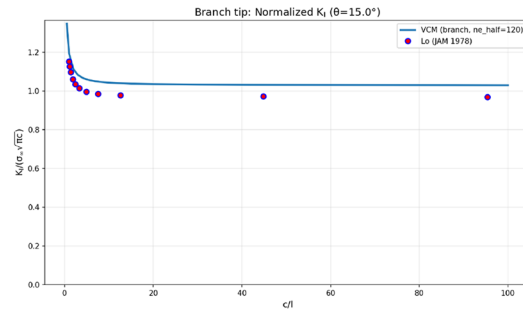
Collectively, these works demonstrate that the analysis of branched cracks primarily rests on Muskhelishvili complex potentials, conformal mapping for geometry reduction, formulations of singular integral equations, perturbation and asymptotic expansions for small branch angles, and numerical quadrature for accurate SIF extraction. The resulting closed-form and semi-analytical SIF solutions provide essential benchmarks for validating modern variational, boundary-element, and finite-element implementations of branched crack mechanics. Here, we compare the numerical solutions to those of Lo (1978). The parameters used in these simulations are as follows: (1) remote stress is $\sigma_{22} = 100$ MPa, (2) $N_{\text{elements}} = 80$ for the symmetric branch and 240 for the asymmetric branch case, (3) the distribution of the element nodes is tip-dense, with a minimum of 3 elements on each side of the kink/junction, a minimum of 6 nodes at the tip, clustering of the tip nodes with a one-sided power of 3 (asymmetric case), or Chebyshev (for the symmetric case), and a minimum of 6 nodes close to the crack tips. Two terms of the Williams field expansion were used to fit the SIF values in the symmetric branch case, and only one term was used for the asymmetric case. When fitting the SIF close to the tip of the crack, the fraction of branch or main crack segments used in the fit was 0.2 for the symmetric case and 0.3 for the asymmetric case, with a minimum number of 8 fitting points. The crack length is $2c$, the branch length is l , and the aspect ratio is c/l .

Figures 5 and 6 compare the current numerical stress intensity factors with the analytical solution of Lo (1978) for a branched crack with a branch angle $\theta = 15^\circ$. The stress intensity factors in mode-I and mode-II are normalized by the remote reference value and are plotted as functions of the branch-length ratio c/l . For the asymmetric configuration, Fig. 5a shows the normalized K_I as a function of c/l . The numerical results reproduce both the magnitude and the strong dependence on the c/l ratio at small values predicted analytically. The most pronounced sensitivity occurs for small c/l , where crack-tip interaction is strongest. Agreement remains close in this regime and improves monotonically as c/l increases.

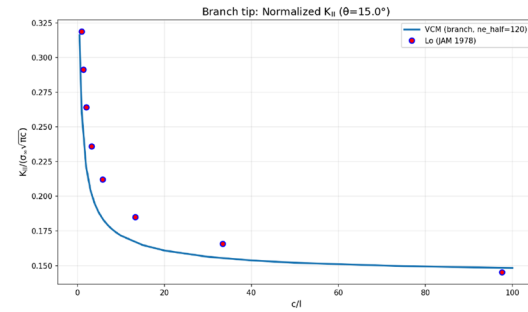
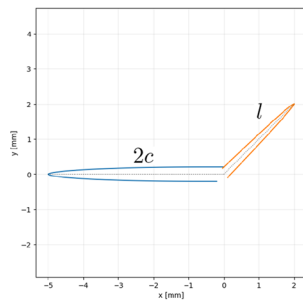
Figure 5b presents the corresponding normalized K_{II} . The shear component decreases rapidly with increasing c/l , consistent with a reduced interaction between the main crack and the secondary branch. The computed values closely match Lo's solution across the full parameter range, with minor deviations only in the small- c/l limit. Figure 6 displays the symmetric configuration. Figure 6a demonstrates excellent agreement in K_I over the entire c/l range, including the interaction-dominated small-branch regime. The solution recovers the correct asymptotic behavior as $c/l \rightarrow 0$ and approaches the single-crack limit for large c/l .

Circular arc crack

Cotterell and Rice (1980) presented a first-order perturbation solution for the stress intensity factors (SIFs) at the tip of a slightly curved or kinked two-dimensional crack. Their work builds on the earlier perturbation approach of Goldstein and Salganik (1974) and employs the complex-variable formulation of Muskhelishvili (1953) for the elasticity

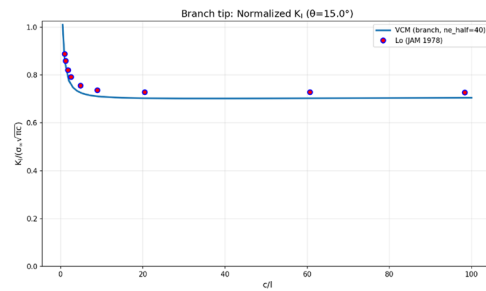


(a) Normalized mode-I stress intensity factor, K_I , versus c/l .

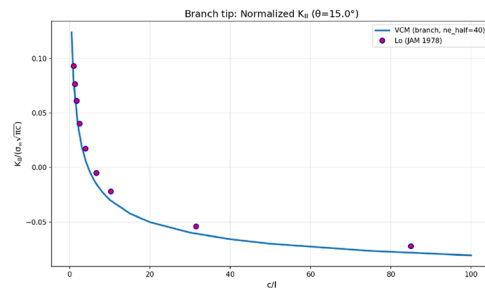
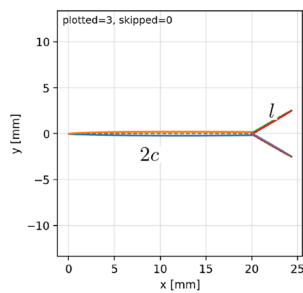


(b) Normalized mode-II stress intensity factor, K_{II} , versus c/l .

Fig. 5 Comparison between the present numerical results and those of Lo (1978) for an asymmetric branched crack. **a** Normalized mode-I stress intensity factors as functions of c/l . **b** Problem geometry. **c** mode-II stress intensity factors as functions of c/l



(a) Normalized mode-I stress intensity factor, K_I , versus c/l .



(b) Normalized mode-II stress intensity factor, K_{II} , versus c/l .

Fig. 6 Comparison between the present numerical results and those of Lo (1978) for a symmetric branched crack. **a** Normalized mode-I stress intensity factor as functions of c/l . **b** Problem geometry. **c** mode-II stress intensity factor as functions of c/l

of the plane. The central idea is to treat a curved crack as a perturbation of a straight crack. Let the crack surface be described by $y = A(x)$, where $A(x)$ is small compared to the crack length. Expansions of the Muskhelishvili analytic functions are written as: $F(z) = F_0(z) + F_1(z) + \mathcal{O}(A^2)$, $W(z) = W_0(z) + W_1(z) + \mathcal{O}(A^2)$, where F_0, W_0 corresponds to the straight crack and F_1, W_1 represents first-order corrections due to curvature. This approach yields closed-form integral expressions for the complex stress intensity factor $K = K_I - iK_{II}$.

Although the primary development in Cotterell and Rice (1980) is a first-order perturbation analysis for slightly curved cracks, the circular arc crack is solved exactly within the complex potential framework. The resulting closed-form expressions for K_I and K_{II} constitute exact solutions for a circular arc crack under remote uniform loading. Closed-form stress intensity factors for a circular arc crack were previously derived by Sih (1970) using the Muskhelishvili complex potential method. Cotterell and Rice (1980) independently re-derived the solution and noted that the published expressions of Sih et al. contained a transcription error in the final algebraic form. The corrected exact expressions are given as (Cotterell and Rice 1980).

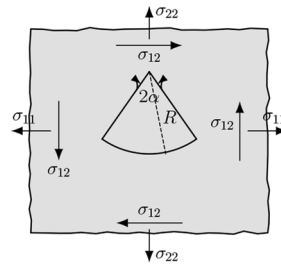
$$K_I = (\pi c)^{1/2} \left\{ \left[\left(\frac{\sigma_{yy} + \sigma_{xx}}{2} \right) - \left(\frac{\sigma_{yy} - \sigma_{xx}}{2} \right) \sin^2 \left(\frac{\alpha}{2} \right) \cos^2 \left(\frac{\alpha}{2} \right) \right] \frac{\cos(\alpha/2)}{1 + \sin^2(\alpha/2)} + \left(\frac{\sigma_{yy} - \sigma_{xx}}{2} \right) \cos \left(\frac{3\alpha}{2} \right) - \sigma_{xy} \left[\sin \left(\frac{3\alpha}{2} \right) + \sin^3 \left(\frac{\alpha}{2} \right) \right] \right\}. \quad (24)$$

$$K_{II} = (\pi c)^{1/2} \left\{ \left[\left(\frac{\sigma_{yy} + \sigma_{xx}}{2} \right) - \left(\frac{\sigma_{yy} - \sigma_{xx}}{2} \right) \sin^2 \left(\frac{\alpha}{2} \right) \cos^2 \left(\frac{\alpha}{2} \right) \right] \frac{\sin(\alpha/2)}{1 + \sin^2(\alpha/2)} + \left(\frac{\sigma_{yy} - \sigma_{xx}}{2} \right) \sin \left(\frac{3\alpha}{2} \right) + \sigma_{xy} \left[\cos \left(\frac{3\alpha}{2} \right) + \cos \left(\frac{\alpha}{2} \right) \sin^2 \left(\frac{\alpha}{2} \right) \right] \right\}. \quad (25)$$

We study here the solution for a circular arc that has a half length (projected on the x-axis) of $a = 10.0$ mm, and vary half the angle of the arc α . For specific fields, we report the results for $\alpha = 30^\circ$. The radius of the arc $R = \frac{a}{\sin(\alpha)}$. For illustration purposes, I used a biaxial remote field $\sigma_{11} = 100$ MPa, and $\sigma_{22} = 200$ MPa.

In Fig. 7, the normal stress field shows classical tensile concentration near the arc tips and compressive shielding beneath the arc, consistent with LEFM asymptotics. The transverse normal stress is redistributed along the arc, with amplification at the tips and symmetry about the vertical axis. However, shear stress exhibits antisymmetric lobes characteristic of mixed-mode contributions induced by curvature. These fields demonstrate that the numerical solution correctly resolves the branch cut along the crack surface.

Figure 8 shows that the crack shape under biaxial loading preserves geometric consistency and symmetry, while the displacement field reveals the square-root singularity structure near the arc tips and confirms correct boundary traction enforcement. Figure 9 displays the RMS error convergence plots and shows that K_I error decreases monotonically with element count, while the K_{II} error reaches a minimum near $N_{\text{half}}^e \approx 100$ and then increases with further refinement, reflecting the competing influence of discretization error and round-off in the SIF extraction at very fine resolution. The mode-I component converges more rapidly, while the mode-II component exhibits slightly greater sensitivity to discretization. In Fig. 10, we compare the numerical SIFs with the analytical predictions of Cotterell and Rice. Excellent agreement is observed across a wide range of



(a) Geometry of the curved arc crack.

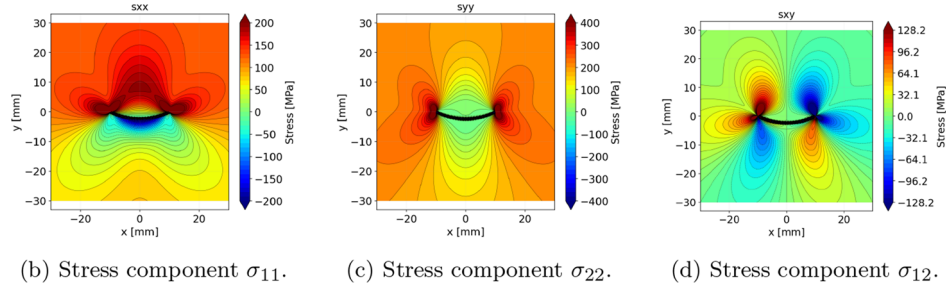


Fig. 7 Stress fields around a curved arc crack. **a** Problem geometry. **b–d** Contours of the stress components σ_{11} , σ_{22} , and σ_{12} . The arc is parametrized as part of a circle with 80 elements, 12 collocation points per element, and 16 Gauss points for field evaluation

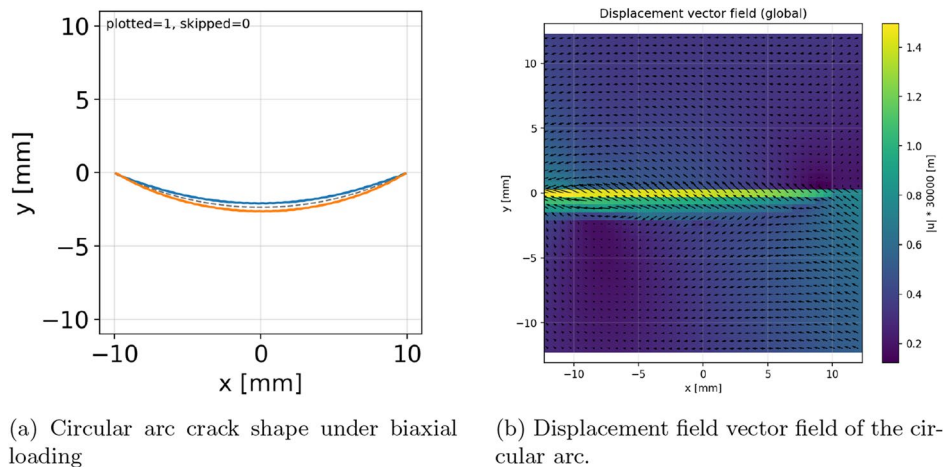


Fig. 8 Deformed shape of the circular arc and the displacement field vector distribution around it. Note the branch cut along the arc itself

arc angles, validating both the curvature representation and the SIF extraction methodology. In general, the validation of the curved crack confirms that the present numerical framework reproduces the analytical solutions of Cotterell and Rice and captures the correct $\sqrt{c}$ scaling and curvature-induced mode coupling. The agreement between these two approaches confirms both geometric fidelity and near-tip field accuracy.

Parametrization of crack networks

Naturally occurring cracks may geometrically resemble polyline geometry with sharp corners, kinks, and branches. However, in some materials, crack networks may be more accurately represented by smooth curves. Such distinction is material and modeling

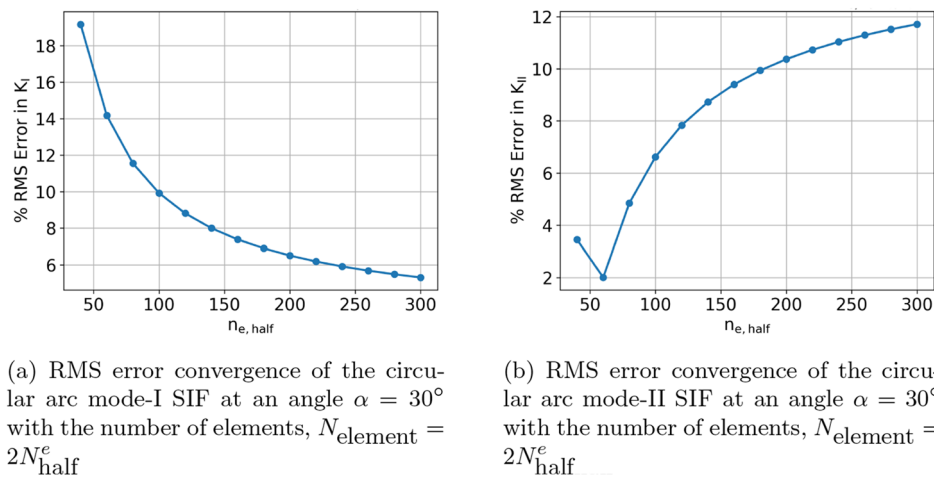


Fig. 9 Convergence tests for the SIF K_I and K_{II}

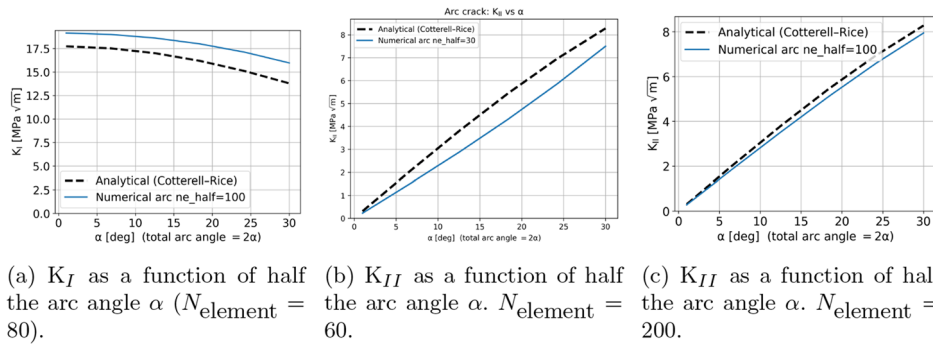


Fig. 10 Stress Intensity factors for a circular arc: a comparison between numerical results and analytical solutions of Cotterell and Rice (1980)

dependent, as the key aspect here is modeling resolution at the small scale. Crack network parametrization smooths localized stress variations resulting from local small discontinuities. The cubic spline eliminates tangent discontinuities, resulting in a smoother redistribution of stress around the network. While the global singular structure at crack tips and junctions remains unchanged, the local stress gradients near former kink locations are attenuated. This smoothing effect reflects the regularization of curvature and the redistribution of elastic interaction over a continuously varying geometry. From a computational perspective, spline parametrization offers important advantages for large crack networks. As networks grow through branching and coalescence, the number of geometric degrees of freedom can increase rapidly. Representing cracks with smooth splines allows curvature to be captured with fewer control points compared to a piecewise-linear discretization. This can significantly reduce the total DOF required to represent complex geometries while preserving the global accuracy of the elastic field. However, the choice of parametrization must be guided by the physical role of geometric sharpness. In cases where kink-induced stress concentrations are physically meaningful—such as in instability or branching studies—the polyline representation retains these localized features explicitly. Conversely, when the focus is on large-scale network evolution or statistical properties of fracture, spline representations may provide improved

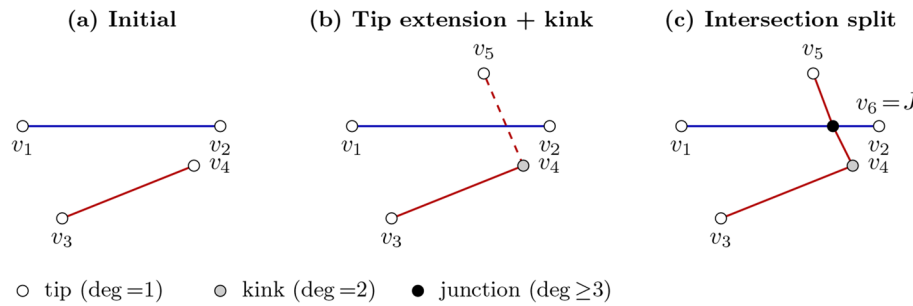


Fig. 11 Topological operations on a two-crack example. **a** Two disjoint cracks, all four vertices are tips ($\text{deg}=1$). **b** Crack B is propagated by appending a new edge at tip v_4 ; the operation $\text{ExtendTip}(v_4, \Delta a, \theta)$ inserts a new tip v_5 and re-classifies v_4 as a kink ($\text{deg}=2$). **c** The new segment (v_4, v_5) crosses edge (v_1, v_2) of crack A; the operation $\text{SplitAtIntersection}$ inserts a junction $v_6 \equiv J$, removes the two crossing edges, and replaces them with four edges meeting at J , which has $\text{deg}=4$

Table 1 Vertex and edge data structure at each stage of Fig. 11. Vertex type is set from graph degree by UpdateVertexTypes , executed after every modifying operation. ExtendTip adds one vertex and one edge; $\text{SplitAtIntersection}$ adds one vertex (the junction J), removes the two crossing edges, and adds four new edges joined at J , preserving connectivity of the rest of the network

Stage	Vertices (id, type)	Edges	Op. producing this stage
(a) Initial	v_1, v_2, v_3, v_4 : tip	$(v_1, v_2), (v_3, v_4)$	—
(b) Extend v_4	v_1, v_2, v_3 : tip; v_4 : kink; v_5 : tip	$(v_1, v_2), (v_3, v_4), (v_4, v_5)$	$\text{ExtendTip}(v_4, \Delta a, \theta)$
(c) Split at J	v_1, v_2, v_3, v_5 : tip; v_4 : kink; v_6 : junction	$(v_1, v_6), (v_6, v_2), (v_4, v_6), (v_6, v_5), (v_3, v_4)$	$\text{SplitAtIntersection}((v_1, v_2), (v_4, v_5))$

numerical efficiency and geometric smoothness without compromising the dominant fracture physics. We examine here various methods of crack network parametrizations.

Topological operations on a two-crack example

Before describing the operations on a general network, we illustrate them on the simplest configuration that exercises every vertex type and every operation: two initially disjoint, non-collinear straight cracks that, as they propagate, develop a kink and intersect. Figure 11 shows three successive stages, and Table 1 lists the underlying vertex/edge data structure at each stage.

A vertex is classified by its graph degree: $\text{deg}(v) = 1 \Rightarrow \text{tip}$, $\text{deg}(v) = 2 \Rightarrow \text{kink}$, $\text{deg}(v) \geq 3 \Rightarrow \text{junction}$, following the assignment in $\text{TopologyManager.update_vertex_types}$. This typing is the only state the propagator needs in order to decide which operations are admissible at each vertex: tips can be extended, kinks can be remeshed, and junctions are kinematically constrained.

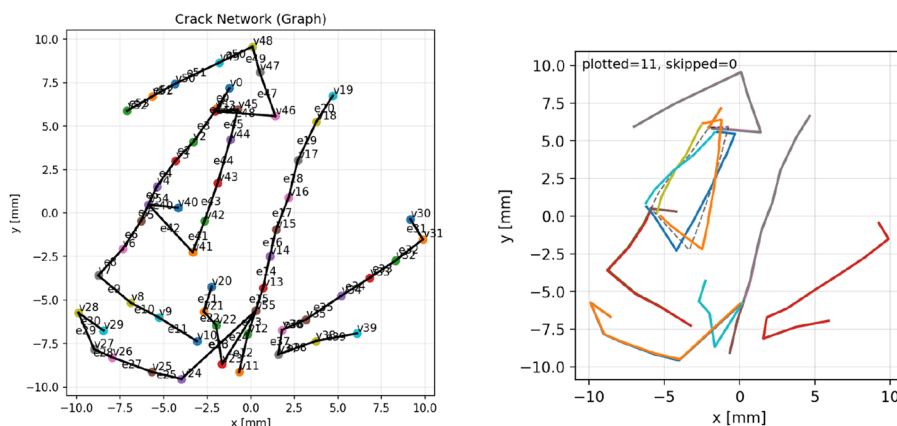
Vertex and edge data structure at each stage of Fig. 11. Vertex type is set from graph degree by UpdateVertexTypes , executed after every modifying operation. ExtendTip adds one vertex and one edge; $\text{SplitAtIntersection}$ adds one vertex (the junction J), removes the two crossing edges, and adds four new edges joined at J , preserving connectivity of the rest of the network.

The general algorithms of the next section (network-level intersection sweep, junction angle filtering, edge proximity check) are direct iterations of the two operations ExtendTip and $\text{SplitAtIntersection}$ illustrated above, applied to all active tips and to all candidate edge pairs, respectively. A reader interested only in the bookkeeping can stop

at this point; the remaining subsections describe how the candidate set is generated and how the operations are scheduled within a single growth increment.

Polyline parametrization

An example of a complex crack network is shown in Fig. 12, where polylines are used for parameterization. The intent of this example is to demonstrate the ability of VCEM to handle a topologically rich network — multiple junctions, multiple kinks, and multiple free tips — in a single static equilibrium solve, as preparation for the later evolving-network simulations of [Fragmentation of compressed disks](#) section. The case is not intended as a physically realistic configuration but rather as a stress test of the topological and constraint-enforcement infrastructure. The network contains internal kinks (degree-2 vertices), free tips (degree-1), and junctions (degree>2), with multiple cracks. The VCEM code first applied topological operations on the network and created a junction node 53 at (−2.05, 5.86) mm, split edge (1, 2) into 1–53 and 53–2, split edge (45, 46) into 45–53 and 53–46, created junction node 54 at (−5.87, 0.49) mm, split edge (4, 5) into 4–54 and 54–5, split edge (40, 41) into 40–54 and 54–41, created junction node 55 at (0.28, −5.62) mm, split edge (12, 13) into 12–55 and 55–13, and split edge (23, 24) into 23–55 and 55–24. As can be seen in the network graph in Fig. 12a. Subsequently, a simulation was carried out on the network with the following parameters: The applied stress ($\sigma_{xx}=50$ MPa, $\sigma_{yy}=100$ MPa), number of elements per polyline=80, shape functions= Chebyshev quadratures, quadrature points =12, Gauss points=16. The deformed shape of the crack network is shown in Fig. 12b. The deformation pattern reveals that crack opening displacement (COD) and crack sliding displacement (CSD) are highly non-uniform across the network: certain segments undergo pronounced opening, while others remain nearly closed. This non-uniformity reflects the combined influence of local orientation relative to the applied stresses (σ_{xx} , σ_{yy}) and global load redistribution mediated through junction compatibility constraints. Because the Burgers densities are determined variationally under these constraints, each branch participates in a



(a) A network graph showing isolated cracks with deg-1 and deg-2 vertices, and cracks joined at several junctions (deg $\geq$ 3).

(b) Corresponding deformed crack network after load application. Note that some segments have much larger COD/CSD than others.

Fig. 12 Illustrative example of a network graph described as a topological object, and the corresponding actual deformed crack surfaces under loading

global equilibrium that couples all segments, even those geometrically remote from one another.

The stress fields shown in Fig. 13 exhibit several distinct features. First, near free tips (deg-1 vertices), the fields display the classical LEFM square-root singularity. In σ_{xx} , σ_{yy} , and σ_{xy} , the contours tighten and form directional lobes aligned with the local crack orientation, consistent with the asymptotic K_I – K_{II} dominated structure. These tip-controlled regions are localized and decay approximately as $r^{-1/2}$ in displacement gradient and r^{-1} in stress magnitude, so their influence diminishes rapidly with distance from the tip. In contrast, internal kinks (deg-2 vertices) do not introduce new singularities but act as curvature-induced field re-orientation. Because the tangent direction changes abruptly, the local Burgers density projection onto the global axes changes sign or magnitude, producing a redistribution of stress lobes across adjacent segments. The stress field therefore exhibits directional rotation across kinks, with contour lines bending to accommodate the change in crack orientation. However, the field remains continuous across the kink, and no additional singular order beyond the crack-tip field is introduced.

Multi-branch junctions (deg ≥ 3 vertices) generate the most distinctive field signatures. Unlike a simple tip, a junction enforces compatibility and Burgers-vector conservation across multiple incident branches. The resulting stress field is not dominated by a single K -field but by the superposition of interacting branch contributions. This produces localized zones of amplified or shielded stress depending on the relative branch orientations and load direction. In Fig. 13, one observes concentrated stress gradients in the vicinity of junction nodes, often accompanied by alternating tensile and compressive sectors. These features reflect constraint-induced redistribution rather than singular tip asymptotics. Junctions therefore act as mechanical coupling hubs, redistributing energy between branches and altering the effective driving force at neighboring tips.

At distances larger than the characteristic network size, the stress field transitions to a coarse-grained, homogenized response. Individual tip singularities and junction-induced oscillations blend into an effective dipolar or multipolar field. The contour plots in Fig. 13 show that far from the network, the field resembles that of an equivalent defect cluster. Thus, while local features are governed by tip singularities and junction compatibility, the far-field behavior is controlled by global equilibrium and the aggregate topological moment of the crack network. In this regime, stress perturbations decay smoothly and align with the principal directions of the applied biaxial loading, indicating

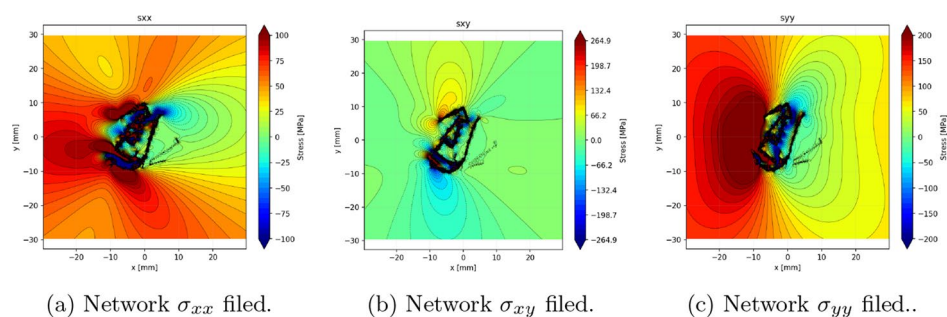


Fig. 13 The stress field of a crack network with multiple junctions, kinks, and crack tips. bi-axially loaded (applied stress $\sigma_{xx}^a = 100$ MPa, and $\sigma_{yy}^a = 200$ MPa)

that the network behaves mechanically as an effective inclusion whose influence extends over scales larger than individual crack segments.

Circular arcs

Figure 14 shows the distribution of σ_{yy} for a circular arc crack subjected to biaxial far-field loading with $\sigma_{xx}^a = 100$ MPa and $\sigma_{yy}^a = 200$ MPa. Although the applied stresses are uniform, the traction-free crack faces redistribute the stress field in a geometry-dependent manner. Along the crack, far-field stresses are projected onto the local tangent-normal frame ($e_1(s), e_2(s)$) as $t_n^a(s) = e_2(s) \cdot \sigma^a e_2(s)$, $t_s^a(s) = e_1(s) \cdot \sigma^a e_2(s)$, where σ^a is the applied stress tensor. Because σ_{yy}^a is twice σ_{xx}^a , the projection onto the local normal direction depends strongly on the orientation of the arc relative to the y -axis. Consequently, even though the remote loading is uniform, the local driving tractions vary continuously along the arc.

For small arc angles (e.g., 15° and 45° in Fig. 9), the crack perturbs the field only locally. The σ_{yy} contours show symmetric lobe structures centered near the tips, with moderate stress relaxation directly beneath the arc. As the arc angle increases (e.g., 90° and 180°), a larger portion of the domain lies within the geometric “shadow” of the crack. Because the crack faces are traction-free, the projected normal traction $t_n^a(s)$ must be relaxed by the induced dislocation field, leading to a pronounced reduction of σ_{yy} in the region enclosed by the arc. This stress relaxation expands spatially as the arc angle grows.

At large arc angles (e.g., 270° and 345°), the crack approaches a nearly closed geometry. The interior region becomes highly compliant, and σ_{yy} is significantly reduced within the enclosed area. The external load is therefore diverted around the arc resembling that around a circular hole, producing intensified stress lobes adjacent to the crack tips and along the outer curvature. These lobes reflect the redistribution of load paths required to satisfy both equilibrium and the traction-free boundary condition.

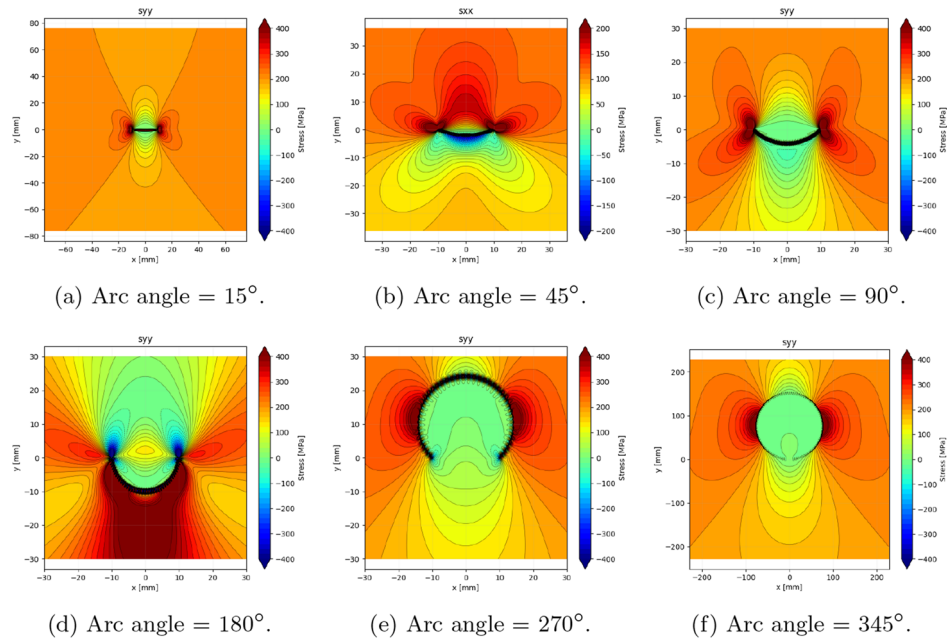


Fig. 14 σ_{yy} for an arc bi-axially loaded (applied stress $\sigma_{xx}^a = 100$ MPa, and $\sigma_{yy}^a = 200$ MPa)

Thus, increasing crack opening increases the effective compliance of the cracked region, producing progressive relaxation of σ_{yy} beneath the arc while concentrating stress outside the shielded region. The evolution of lobe shapes with arc angle, therefore, provides a field-level manifestation of curvature-induced mode coupling and geometry-controlled stress redistribution under biaxial loading.

Cubic splines

Figures 15 and 16 illustrate the flexibility of the VCEM framework in parametrizing complex crack networks and the sensitivity of the resulting elastic fields to geometric representation. As shown in Fig. 15, the network consists of a branched configuration containing two degree-3 junctions and a geometric kink. Two distinct geometric parametrizations are considered: (i) a piecewise-linear polyline representation and (ii) a smooth cubic spline representation of the same network topology.

In the polyline representation (Fig. 15b), each crack segment is modeled as a straight element with a discontinuous tangent at kinks. This parametrization captures geometric sharpness explicitly and preserves localized angular changes in crack direction. In contrast, the cubic spline representation (Fig. 15c) enforces C^1 continuity of the crack centerline, smoothing curvature while preserving the same junction connectivity. The spline representation reduces geometric discontinuities without altering the topological structure of the network.

The influence of parametrization becomes particularly evident in the stress fields shown in Fig. 16. For the polyline representation, the stress components σ_{xy} and σ_{xx} exhibit localized angular features near the kinked regions. These features arise from abrupt changes in crack tangent direction, which introduce localized variations in traction projection and elastic interaction kernels. The stress contours display sharp directional transitions near kinks, consistent with the geometric discontinuity in curvature. These results demonstrate that VCEM decouples topological structure from geometric representation: the underlying variational formulation and kernel operators remain unchanged, while the parametrization strategy controls geometric smoothness, stress localization, and computational cost.

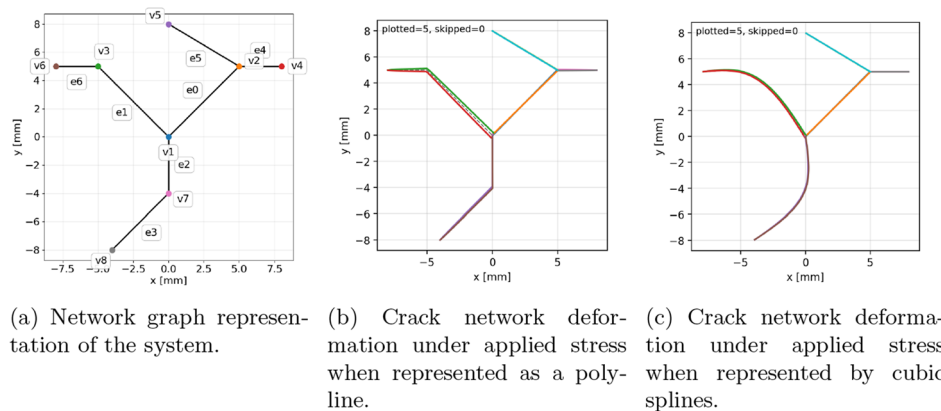


Fig. 15 Illustration of a crack that has branched in two degree-3 junctions, with one kink. Two realizations are shown: the network is parametrized with straight polylines, and (2) the network is parametrized with cubic splines

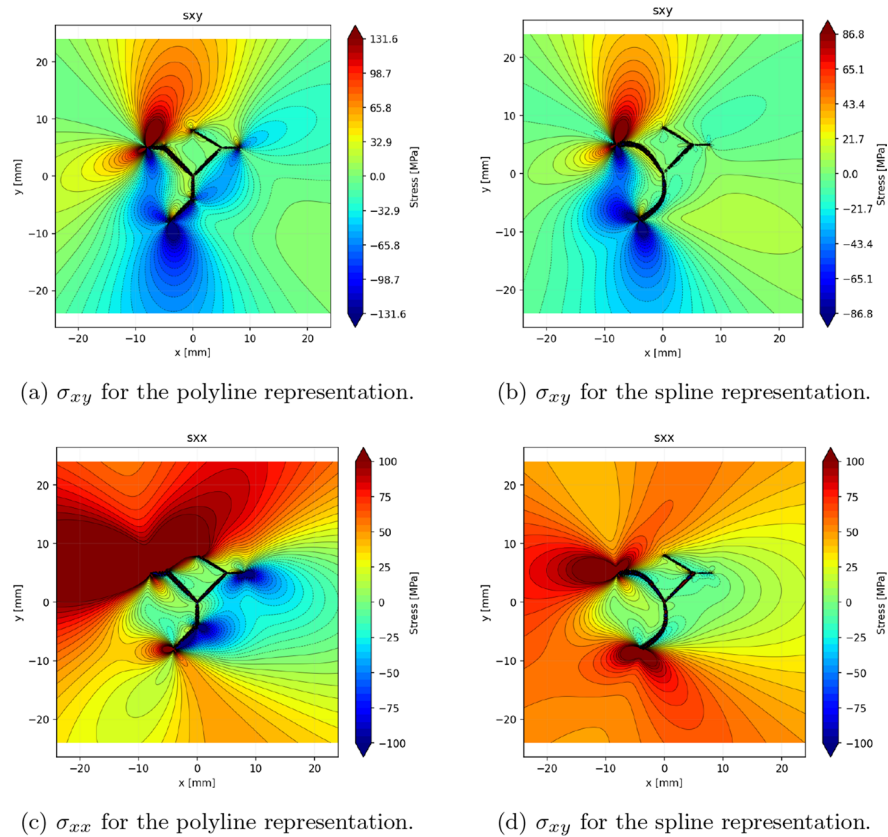


Fig. 16 Stress field components of the network. The network is represented by straight polylines and by cubic splines to reveal the effects of geometric representation on the structure of the elastic field

Simulations of disk compression experiments

Background and motivation

Diametral compression of circular disks, commonly referred to as the Brazilian disk test, has long served as a fundamental experimental configuration for probing tensile fracture under globally compressive loading conditions. The classical elastic solution of Hondros Hondros (1959) established the internal stress distribution in a diametrically loaded disk, demonstrating that a tensile stress field develops along the loaded diameter despite externally compressive boundary tractions. This counterintuitive feature renders the geometry uniquely suitable for studying tensile fracture initiation and mixed-mode crack propagation in brittle solids. Subsequent analytical developments by Atkinson and co-workers (Atkinson et al. 1973; Atkinson and Gross 1977) extended the formulation to include inclined cracks and mixed-mode stress intensity factors, while the experimental and theoretical investigations of Shetty et al. (1987) provided detailed measurements of crack initiation angles and mixed-mode fracture criteria in the disk configuration.

A distinctive feature of the Brazilian disk is its sensitivity to the loading protocol. Under force-controlled loading (constant applied load or pressure), crack extension increases specimen compliance, which in turn amplifies the driving force for further propagation, often producing unstable or catastrophic fracture. In contrast, displacement-controlled loading can stabilize crack growth and enable the measurement of resistance curves. This dual character allows the configuration to probe both force-controlled and displacement-controlled failure regimes within a single geometric setting.

The disk compression test is widely employed to determine indirect tensile strength, mode-I fracture toughness, mixed-mode fracture envelopes, and crack initiation criteria, particularly in rocks, ceramics, glasses, and geomaterials for which direct tensile testing is experimentally challenging (International Society for Rock Mechanics 1978, 2007). Compared with four-point bending or compact-tension specimens, the disk geometry offers several advantages: minimal machining and gripping requirements, geometric simplicity, symmetry that facilitates analytical and numerical modeling, and the natural presence of biaxial stress components relevant to geomechanics and multiaxial failure. Because fracture occurs within an overall compressive environment, the Brazilian disk also provides valuable insight into crack-tip behavior under confinement and the transition between tensile opening and compression-dominated shear, phenomena of central importance in rock mechanics and brittle failure under multiaxial stress states. For these reasons, the Brazilian disk has become a canonical benchmark for validating fracture criteria, mixed-mode theories, and computational methods in finite elastic domains.

The fracture behavior of a compressed alumina disk is examined here. It is schematically shown in Fig. 17, with simulation parameters corresponding to our experiments (Alabdullah et al. 2023), as follows: elastic modulus $E = 231.5$ GPa, Poisson's ratio $\nu = 0.3$, disk diameter = 25.4 mm, disk thickness = 6.35 mm, number of BEM elements = 120, number of crack elements = 240, applied force 3.83 kN, BEM Gauss points = 6, crack shape functions = Chebyshev quadratures, node distribution = tip dense, solver option = parametrized crack, parametrization = polyline, collocation points = 9, Gauss points = 12, and crack mode = polar.

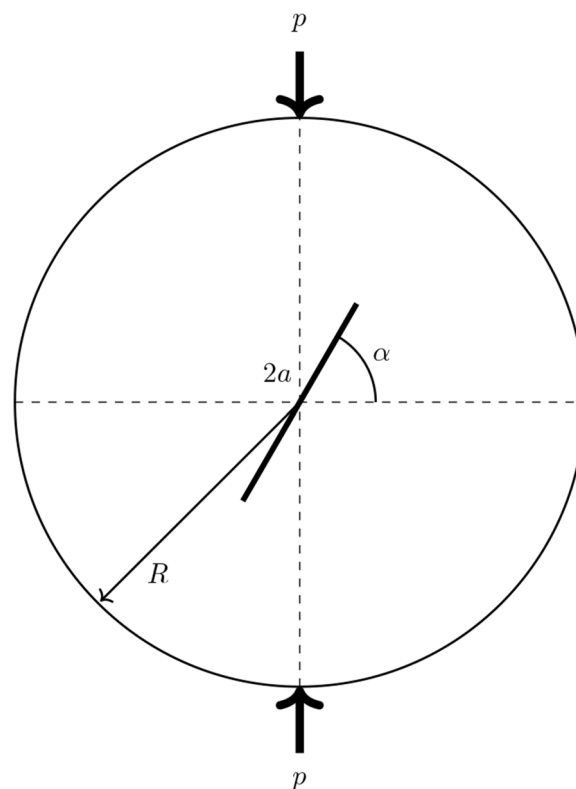


Fig. 17 Geometry of a diametrically compressed disk with an inclined center crack of length $2a$

Crack growth algorithm

Crack evolution is treated incrementally as a sequence of interaction-aware equilibrium solves followed by local, per-tip geometric updates. The key structural feature of the algorithm is that *crack interactions are resolved at the equilibrium step, not at the growth step*. Every active tip sees the Burgers densities of every other crack through the Kelvin-based kernel (Eq. (2)) in the KKT solve (4), so the (K_I, K_{II}) values on which the per-tip growth rule operates already contain shielding, amplification, and mode-mixity induced by the entire network. The growth rule itself is then applied locally at each tip, but on a globally informed field. No tip is advanced on a field computed in isolation, and all active tips are advanced simultaneously within a single cycle rather than sequentially, in order to avoid any ordering bias in the presence of multiple comparably active tips.

For a crack configuration $\Gamma^{(n)}$ at cycle n , the following sub-steps are performed:

1. **Interaction-aware equilibrium solve.** The constrained variational problem (4) (or its BEM-coupled form (5) in bounded domains) is solved to obtain the Burgers density $b^{(n)}(s)$ on every crack in the network. Because the stiffness block H couples every DOF on every crack and the junction-closure constraints $C_{JQ} = 0$ together with the unilateral non-penetration constraints are enforced *globally*, $b^{(n)}$ is the fully interacting equilibrium solution.
2. **Active-tip inventory and near-tip field extraction.** Let $\mathcal{T}^{(n)} = \{\tau_1, \dots, \tau_{N_t}\}$ be the set of degree-one vertices of the current network graph. At every tip $\tau_i \in \mathcal{T}^{(n)}$, the SIF components $(K_I^{(i)}, K_{II}^{(i)})$ are extracted from the converged solution using the procedures of Sec. 3.1.1. Extraction is performed tip-by-tip, but the input fields are the converged interacting fields from Step 1.
3. **Activity screening.** At each tip the effective SIF $K_{\text{eff}}^{(i)} = \sqrt{(K_I^{(i)})^2 + (K_{II}^{(i)})^2}$ is compared to the local fracture toughness:

$$\text{tip } \tau_i \text{ active} \iff K_{\text{eff}}^{(i)} \geq K_c(x_{\tau_i}), \quad i = 1, \dots, N_t, \quad (26)$$

where $K_c(x)$ is either a constant (deterministic case) or a sampled realization of a spatial random field (microstructure-aware case; see Sec. 5.3 of our paper (Ghoniem 2026a)). Tips with $K_{\text{eff}}^{(i)} < K_c$ remain stationary during the current cycle. Tips entering a compressive confinement zone, for which $K_I^{(i)} \leq 0$, are also held stationary: tensile extension is energetically inadmissible, and any further kinematic response is carried by the unilateral contact constraint (Sec. 4.5 of our paper (Ghoniem 2026a)) rather than by tip advance. This mechanism accounts for the arrest of the centerline-crack tips in Fig. 18 as they penetrate the compressive zone beneath the loading arcs.

4. **Per-tip kink angle.** For every active tip, the propagation angle θ_i , measured from the tangent of the parent segment, is computed from the converged mixed-mode SIFs at that tip using a standard local criterion. The default is the maximum-hoop-stress (MTS) criterion of Erdogan–Sih (Erdogan and Sih (1963)),

$$\theta_i = 2 \arctan \left[\frac{1}{4} \left(\frac{K_I^{(i)}}{K_{II}^{(i)}} \pm \sqrt{(K_I^{(i)}/K_{II}^{(i)})^2 + 8} \right) \right], \quad (27)$$

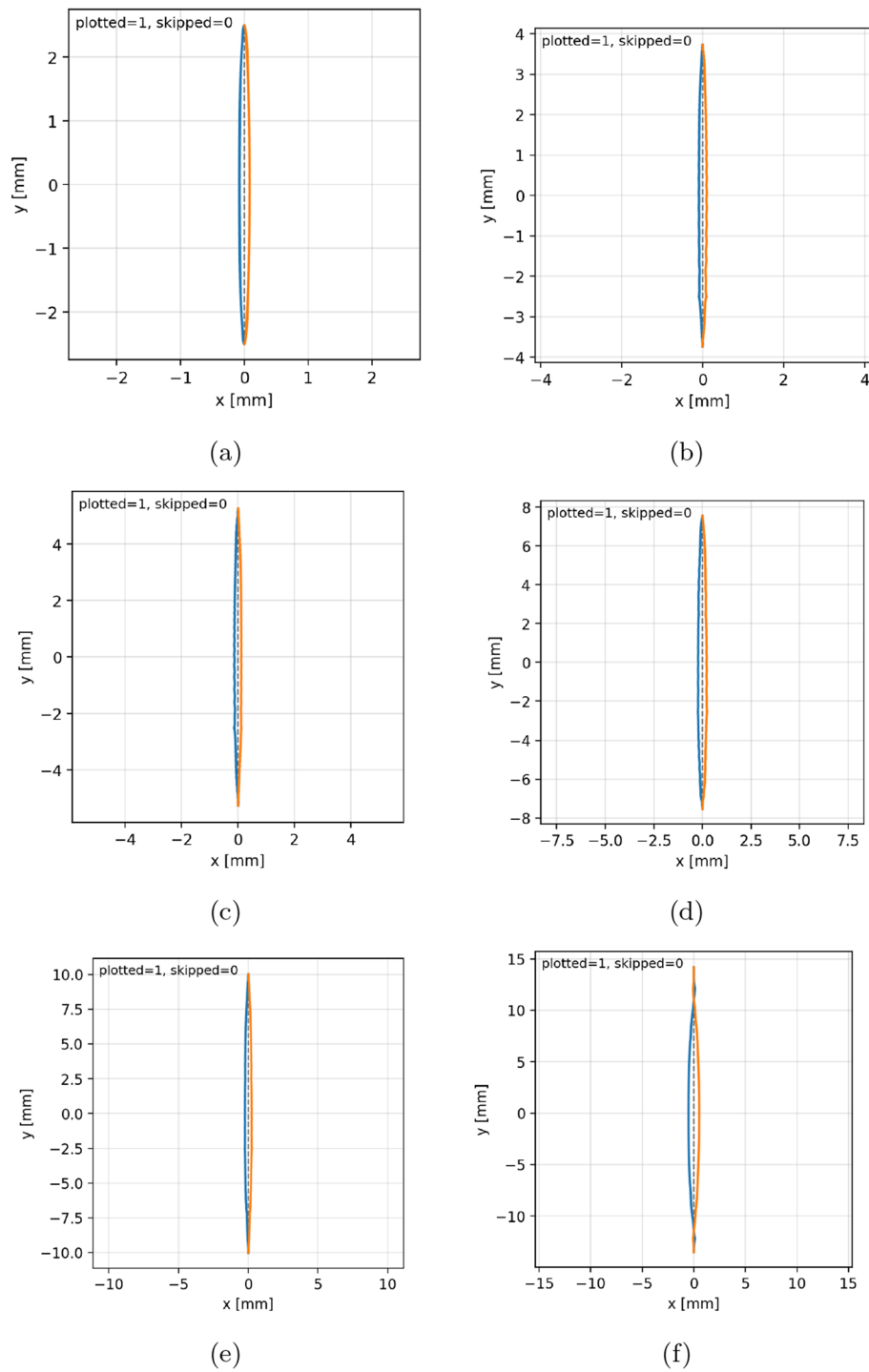


Fig. 18 Growth stages of a centerline crack in the compressed disk. Simulations were performed with the iterative BEM-KKT solver. Note that the two tips “sharpen” in **e** and totally close in **f** as the tips penetrate through the compressive zone under the loading arc

with the sign chosen so that $\sigma_{\theta\theta}$ is maximised along θ_i (equivalently, so that $|K_{II}|$ at the next cycle is reduced). The local-symmetry criterion $K_{II}(\theta_i) = 0$ and the strain-energy-density criterion of Sih Sih (1974) are also available as drop-in alternatives (Sec. 5.2 of our paper (Ghoniem 2026a)). The absolute propagation direction of the new segment at tip τ_i is

$$\varphi_i = \varphi_{\text{tangent}}^{(i)} + \theta_i, \quad (28)$$

where $\varphi_{\text{tangent}}^{(i)}$ is the global orientation of the parent segment at tip τ_i . Each θ_i is therefore determined from the $(K_I^{(i)}, K_{II}^{(i)})$ pair produced by the fully interacting solve, so the kink angle at tip τ_i already contains the shielding and amplification induced by all other cracks.

5. **Per-tip propagation increment.** The propagation increment at tip τ_i is chosen as a fixed fraction of a local length scale,

$$\Delta a_i = \eta \ell_i^{\text{ref}}, \quad \eta \in (0, 1], \quad (29)$$

where ℓ_i^{ref} is the length of the parent segment at τ_i (or a user-prescribed reference length, whichever is smaller) and η is a global step-size parameter held constant throughout a simulation. As a result, Δa_i scales automatically with the local discretization and no tip advances through more than one local length scale in a single cycle, which prevents the growth rule from biasing the evolution towards the fastest-growing tip. As discussed in Sec. 5 of reference (Ghoniem 2026a), η should be small enough that the near-tip fields used to set θ_i remain locally representative over the extension; convergence with respect to η is checked by repeating selected simulations with $\eta/2$ and confirming that the predicted paths are insensitive to the step.

6. **Simultaneous geometric update and topology check.** All active tips are advanced *simultaneously*: a new segment e_i^{new} of length Δa_i and absolute direction φ_i is appended at each active τ_i , and the network graph is updated in a single pass (Sec. 2 of our paper (Ghoniem 2026a)). Two events are detected automatically during this update: (i) if a newly extended segment intersects another crack branch, a degree-3 (or higher) junction is inserted at the intersection point, the intersected edges are split, and the junction-closure constraints $C_{Jq} = 0$ are re-assembled on the updated graph; (ii) if an extended segment reaches the outer boundary $\partial\Omega$, the corresponding tip is removed from $\mathcal{T}$ and a boundary-terminated polar crack is registered through the BEM coupling operators of Eq. (5).
7. **Re-equilibration.** After the geometric update, the KKT system (4) (or (5)) is reassembled on $\Gamma^{(n+1)}$ and re-solved. Shielding or amplification induced by the new segments is therefore immediately reflected in the Burgers density and in the per-tip SIFs used at the next cycle's Step 2, closing the interaction loop.

Centerline slit: growth, closure, and shear localization

Figures 18 and 19 show the quasi-static growth of a centerline slit in a Brazilian-disk geometry under load control. The disk is subjected to constant pressure applied over two opposing arcs of 15° each ($\alpha = 90^\circ$ in Fig. 17). The loading produces the classical non-uniform stress state of the compressed disk: strong compressive zones under the loading arcs and an interior field that redistributes forces through the bulk. As the slit advances along the centerline, the crack remains geometrically aligned with the disk diameter, but its *mechanical character* changes markedly as the tips approach the compressive contact zones.

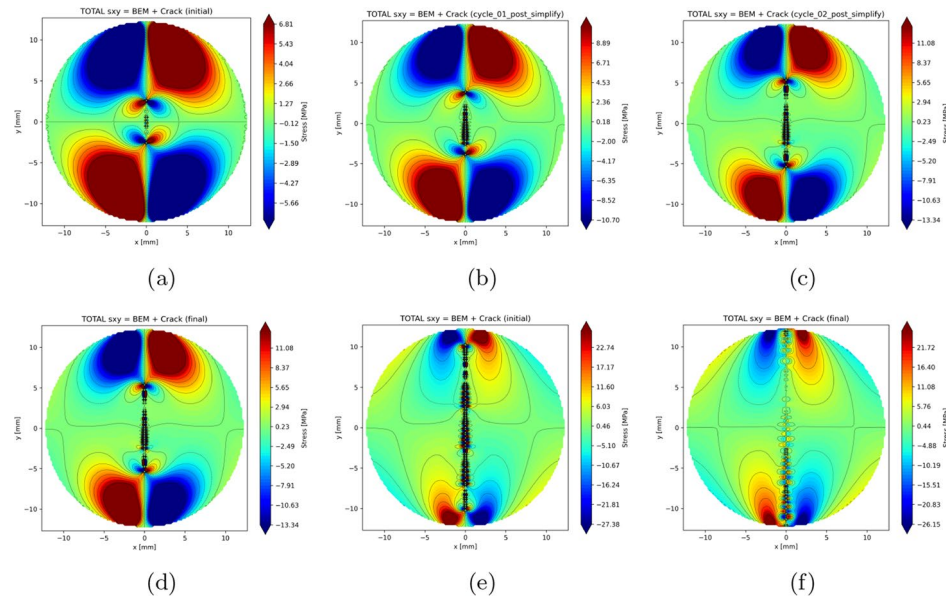


Fig. 19 σ_{xy} in the compressed disk at various stages of growth

The growth sequence in Fig. 18a–d is consistent with a tensile-dominated splitting process in the interior: the crack provides a traction-free path that relaxes tensile stress components and concentrates the driving field at the tips. However, the later stages exhibit a qualitative transition. In stage (e), the crack tip visibly sharpens as it advances into the near-contact region where the normal stress becomes increasingly compressive. In this regime, the local mode-I intensity is reduced toward zero and may become negative ($K_I \lesssim 0$), indicating that the crack faces are driven into contact rather than opened. In stage (f), this effect becomes explicit: the crack tips exhibit clear closure as they enter the compression-dominated region under the loading arcs. Physically, this signifies that the crack can no longer be treated as fully traction-free near the tip; instead, the local mechanics transition from opening-dominated fracture to a closure/contact-controlled state in which compressive normal tractions suppress the crack opening displacement and diminish the energetic driving for further extension.

The corresponding evolution of σ_{xy} is shown in Fig. 19. At early stages, the shear field exhibits a relatively broad structure in the interior, reflecting global load transfer around the slit and the nonlocal nature of stress redistribution in a finite disk. As the crack grows, the σ_{xy} contours become progressively more concentrated along the crack path and near the tips. This localization has two complementary physical origins. First, crack extension increases the disk compliance and forces the applied tractions to be carried by an increasingly constrained ligament, thereby intensifying stress gradients near the advancing tip. In the absence of a through-going separation, the load path must “flow around” the crack, producing enhanced shear in the surrounding material and concentrating σ_{xy} where the elastic field must accommodate sharp changes in boundary conditions (traction-free faces versus compressed boundary arcs). Second, as the tip enters the compressive zone and K_I is driven toward zero and then negative, the crack faces partially close. Closure suppresses tensile opening but does *not* eliminate tangential tractions: instead, shear transfer becomes increasingly important. In a physical disk, this would be accompanied by contact/frictional constraints along closed crack faces, which

promote shear traction transmission and further concentrate the σ_{xy} field. Thus, the observed sharpening of σ_{xy} near the later stages of growth is interpreted as a transition from a predominantly opening-driven interior response to a closure-controlled near-contact response, where shear is confined to a narrower zone as the tip approaches the boundary and the remaining ligament is forced to carry the applied load.

These results emphasize that crack growth in diametral compression is not governed by a single, uniform fracture mode along the entire trajectory. A centerline splitting crack can be tensile-driven in the interior while becoming closure-dominated near the loading contacts. The late-stage tip sharpening and subsequent closure observed in Fig. 18e–f therefore mark a mechanically meaningful crossover: the crack advances into a region where compressive confinement suppresses mode-I opening, alters the local asymptotics, and promotes localized shear transfer, as evidenced by Fig. 19. From a modeling standpoint, this crossover highlights the importance of incorporating crack-face contact effects when interpreting near-contact propagation and the terminal stages of splitting in Brazilian-disk configurations.

Compressed disk with an off-axis slit

Figures 20 and 21 illustrate the evolution of an initially inclined slit in a diametrically compressed disk ($\alpha = 60^\circ$ in Fig. 17), computed using the iterative BEM–KKT solver. The crack is initially oriented at a finite angle with respect to the compression axis, thereby inducing a mixed-mode stress state at the crack tips. In this configuration, both K_I and K_{II} are nonzero, and the initial propagation direction is governed by the local angular distribution of the crack-tip hoop stress. At the onset of loading, the inclined slit experiences a biaxial stress state characteristic of the Brazilian disk geometry. Although the external loading is compressive, the interior of the disk develops tensile stresses transverse to the loading direction. The inclined crack, therefore, experiences a

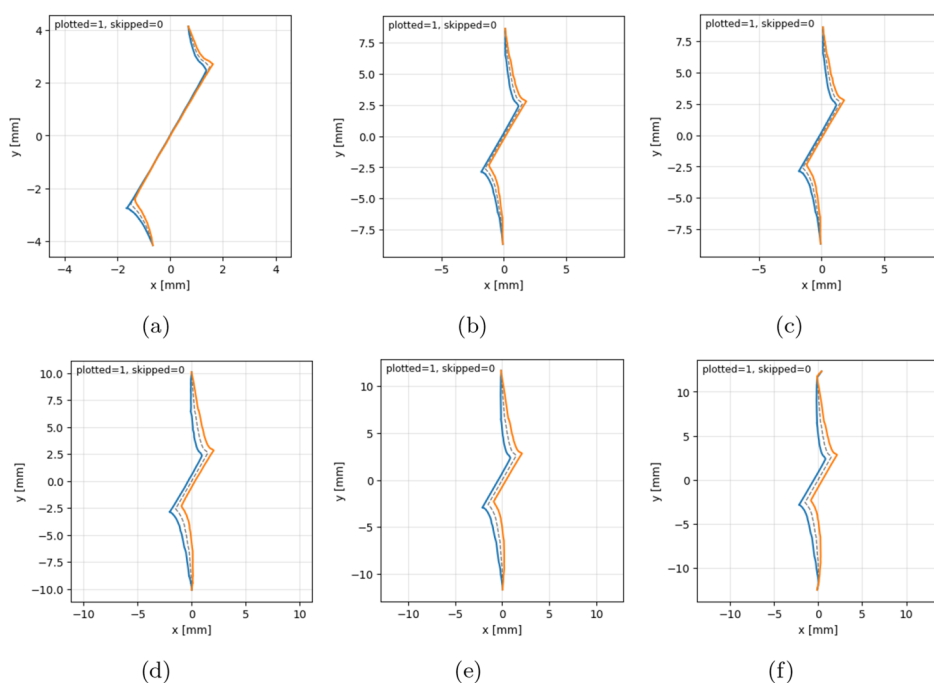


Fig. 20 Inclined crack growth in the compressed disk. Simulations were performed with the iterative BEM–KKT solver. Notice the change in scale as the crack grows from its initial configuration in **a** to its final configuration in **f**

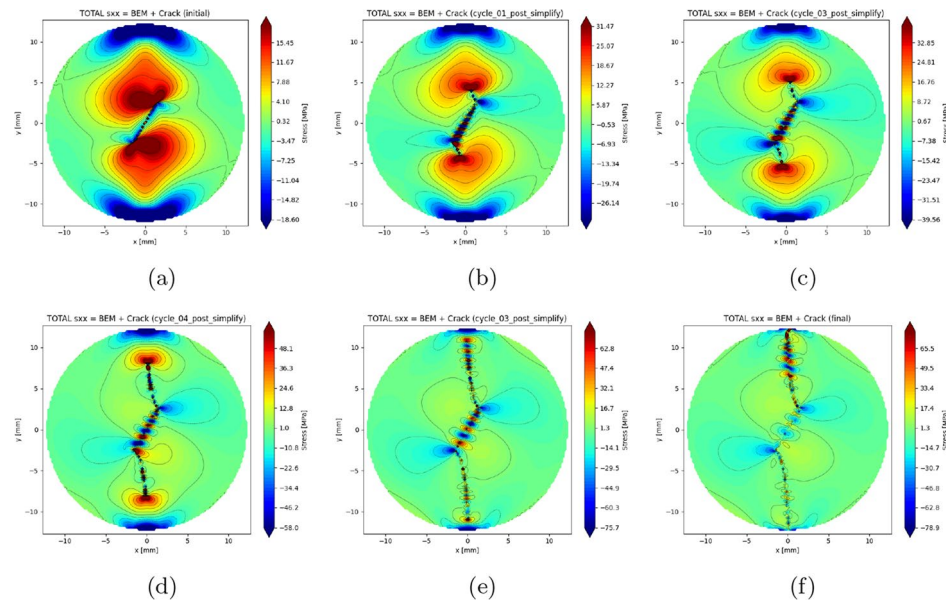


Fig. 21 σ_{xx} in the compressed disk at: **a** the initial state, **b** through **f** after crack growth

mixed-mode state: the shear component K_{II} arises from geometric misalignment with the principal tension direction, while K_I is associated with the tensile opening of the crack tips. Consistent with maximum hoop stress arguments, the crack first kinks non-coplanarly, forming a steeper segment relative to the original slit orientation. This initial deflection reflects the angular location of maximum tensile hoop stress around the tip under combined K_I – K_{II} loading.

As the crack extends, the mode mixity decreases. The shear component is progressively relaxed through geometric reorientation, and the crack path curves toward the vertical diameter of the disk. In the Brazilian geometry, the vertical axis corresponds to the direction of maximum energy release under diametral compression. Once the crack approaches this axis, the loading becomes effectively pure mode I, and further growth proceeds in a symmetric splitting configuration. The final morphology, therefore, corresponds to the classical tensile splitting failure mode of the compressed disk. The evolution of the stress field is shown in Fig. 21 through contours of σ_{xx} . In the initial state (Fig. 21a), the uncracked disk exhibits the well-known biaxial pattern: compressive lobes near the loading contacts and a central tensile region. Upon crack initiation (Fig. 21b), tensile stresses concentrate at the advancing crack tips, while adjacent regions experience stress relaxation due to the formation of traction-free surfaces. The crack acts as a stress sink, redistributing internal forces by diverting compressive load paths around the growing discontinuity. As growth proceeds (Fig. 21c through Fig. 21f), tensile stresses become highly localized in a narrow vertical band aligned with the crack, whereas the surrounding disk is increasingly dominated by compression. This localization reflects the progressive focusing of elastic energy into the remaining ligament ahead of the crack tip. A notable feature of the diametral-compression configuration is that fracture occurs within a globally compressive structure. The tensile crack-tip fields are embedded within a surrounding compressive stress domain. Behind the crack tip, the faces are subjected predominantly to compression, which suppresses shear sliding and reduces the effective

K_{II} contribution. Ahead of the tip, however, the local tensile hoop stress remains positive and drives propagation.

The coexistence of tensile tip fields and compressive lateral zones produces a self-stabilizing mechanism: shear components are progressively eliminated, and the crack aligns with the principal tensile direction. Thus, the vertical splitting configuration emerges as the energetically preferred trajectory under diametral compression. The detailed evolution of this process is quantified in Fig. 22, where the stress intensity factors K_I and K_{II} are plotted as functions of the crack-tip position along the loading axis. At early stages, both components are nonzero, reflecting a mixed-mode state induced by geometric misalignment between the crack and the tensile axis of the disk. As the crack advances, the tip systematically reorients to maximize the opening component, driving K_I toward a local maximum while rendering K_{II} progressively smaller. This reorientation is consistent with the maximum circumferential stress criterion and reflects the tendency of the system to reduce shear-driven asymmetry in favor of tensile opening.

The non-monotonic behavior of K_I in Fig. 22 reflects three geometric regimes that the inclined-slit tip traverses as it propagates along the loading axis of the diametrically compressed disk (shaded in Fig. 22):

- $4 \text{ mm} \lesssim y \lesssim 8 \text{ mm}$ — the *tensile-dominated* regime in which the remote stress at the tip is the near-uniform Brazilian-disk tension; K_I grows slowly with crack length and $K_{II} \approx 0$ because the maximum-hoop-stress direction law repeatedly chooses the kink angle that nullifies K_{II} .
- $8 \text{ mm} \lesssim y \lesssim 11 \text{ mm}$ — the *geometric stress-concentration band* where the tip approaches the curvature of the disk boundary and the local $\sigma_{\theta\theta}$ at the tip is amplified by interaction with the curved free boundary; this produces the peak in K_I near $y \approx 9.5 \text{ mm}$. K_{II} remains within the angle-tolerance band $|K_{II}| \lesssim K_I \tan(\text{tol}_{\theta, \text{abs}})$ of the propagation step controller and is therefore not increasing in any physically meaningful sense.

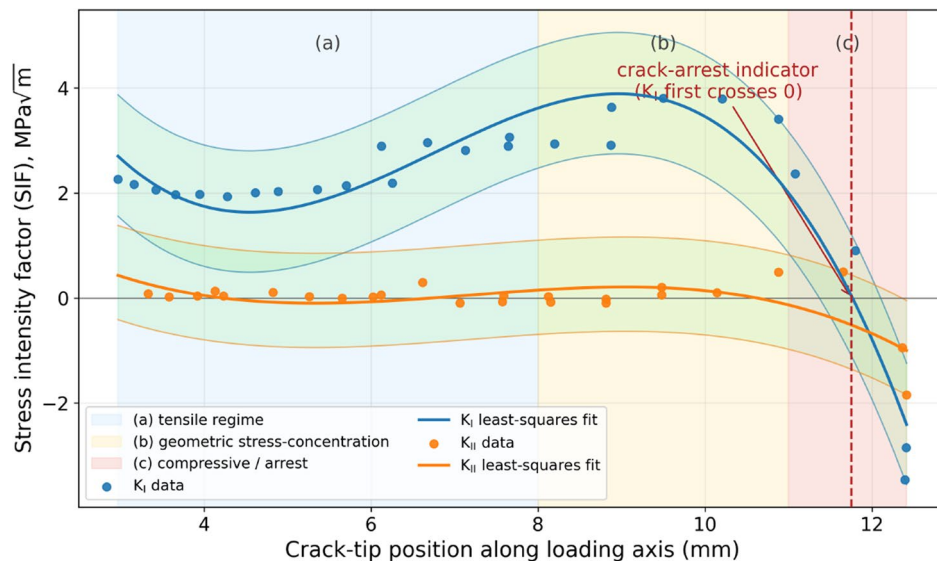


Fig. 22 Stress intensity factors, K_I and K_{II} at the tip of the advancing cracks in the compressed disk as functions of the tip position along the loading axis

- (c) $y \gtrsim 11$ mm — the tip enters the compressive Hertzian zone directly underneath the loading arc; σ_{yy} at the tip changes sign and the computed K_I drops sharply, crossing zero near $y \approx 12$ mm (*crack-arrest indicator* marked in Fig. 22). A negative computed K_I does not correspond to physical mode-I propagation: it would require crack face interpenetration, which is precluded by the unilateral contact constraint at the crack faces in any physically-realizable loading. We retain these last cycles in the plot for completeness, but they are to be interpreted as a crack-arrest signal rather than as continued growth.

The simulation enforces the propagation criterion $K_{\text{eff}} \geq K_c$ at every cycle (Circular arcs section), and the maximum-hoop-stress direction law selects the kink angle that drives $K_{II} \rightarrow 0$ along the propagation path; both criteria are therefore satisfied at every plotted marker. The non-monotonic K_I seen in regimes (b)–(c) is a geometric consequence of the disk–contact stress field, not a bypass of the SIF-based criterion.

As the crack approaches the compression-dominated regions beneath the loading arcs, the local normal stress field changes character. The confining stress increases, the crack-tip opening displacement diminishes, and K_I decreases toward zero and may become negative, indicating incipient crack-face closure. Simultaneously, K_{II} is reduced and may also change sign as the crack enters a region where shear transfer is constrained by compressive confinement. Near the disk boundary, both K_I and K_{II} can become negative, signaling that the tip has transitioned into a regime where compressive stresses dominate the near-tip asymptotics and tensile driving is suppressed. The crack tip must therefore rotate away from the load axis as closure develops, reflecting a local loss of tensile admissibility (see Fig. 20f). Physically, this marks the crossover from an interior tensile-dominated growth regime to a boundary-controlled confinement regime.

Viewed from an energetic perspective, the non-monotonic evolution of K_I and K_{II} reflects an underlying variation in the energy release rate $G = -d\Phi/dA$ along the crack path. In the interior of the disk, crack extension reduces the total potential energy by relieving tensile stresses concentrated along the splitting diameter. As the crack nears the boundary, however, further extension yields diminishing energetic benefits because the advancing tip enters a compressive field that suppresses opening and reduces the driving force. The change in sign of the stress intensity factors, therefore, corresponds to a local reversal of the energetic descent direction in configuration space. In this sense, crack growth in the compressed disk can be interpreted as motion on an evolving energy landscape: the crack initially follows a descending path toward a tensile-aligned configuration, but upon reaching the confinement-dominated region, the landscape flattens and eventually inverts. This interplay between tensile energy release and compressive stabilization provides a natural bridge to the interpretation of fracture as a phase transition, where crack extension corresponds to transitions between metastable elastic states of differing topology. The interior growth stage resembles a symmetry-breaking process in which the crack aligns with the principal tensile axis, whereas the boundary-induced closure regime illustrates how confinement can arrest or redirect the transition. The Brazilian disk, therefore, offers a controlled setting in which the energetics of crack reorientation, instability, and confinement can be interpreted within a unified phase-transition framework.

The comparison in Fig. 23 shows the predicted first-segment kink angle θ (from the present simulations using a maximum hoop-stress criterion) plotted against the initial

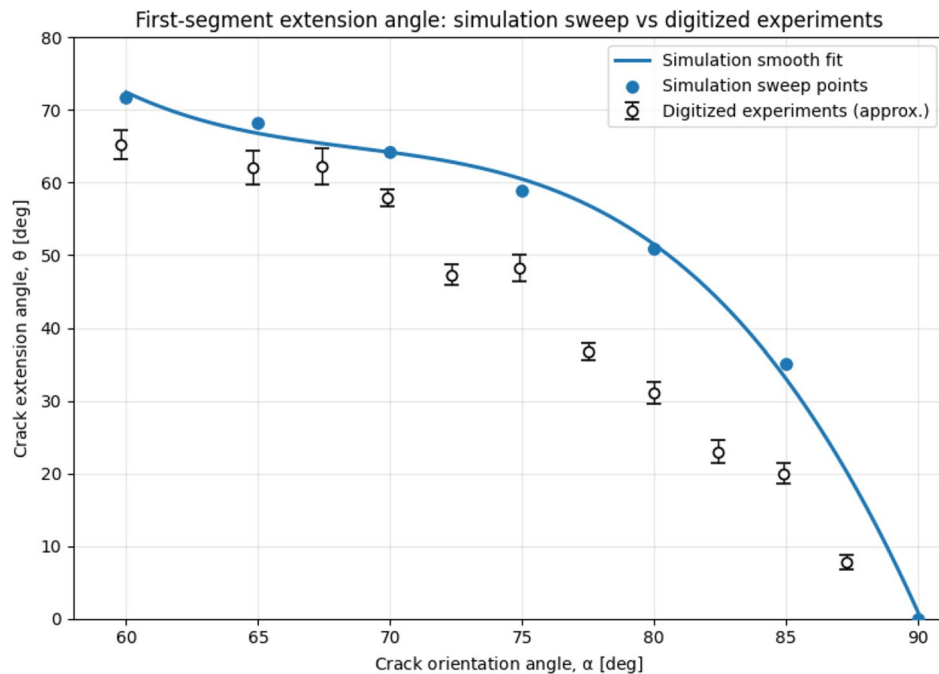


Fig. 23 A comparison between numerical simulations and experimental data on the initial crack growth angle relative to the crack direction α (Shetty et al. 1987)

crack orientation α , together with digitized values from the experiments of Shetty and co-workers (Shetty et al. 1987). Overall, the simulations reproduce the correct trend: as α increases toward the nominal mode-I configuration ($\alpha \rightarrow 90^\circ$), the kink angle decreases rapidly and approaches $\theta \approx 0$, while for smaller α the crack kinks more strongly. The agreement is qualitatively good; however, the simulations systematically predict larger kink angles than the experiments across most of the range. Several physically plausible mechanisms can account for this consistent offset. First, the experimental configuration likely involves partial crack-face contact and friction, especially for orientations where the normal traction on the crack is compressive or near-zero; such contact suppresses relative sliding and effectively reduces the operative mode-II driving, i.e., it diminishes the effective K_{II} transmitted to the tip. Since the maximum hoop-stress criterion maps mode mixity into a kink angle, reducing the effective shear component leads directly to smaller observed θ than a traction-free LEFM prediction. Second, for α near 90° (nominally mode I), the crack-tip state is highly sensitive to small departures from ideal traction-free conditions: compressive confinement beneath the loading arcs and finite contact width can promote crack closure and reduce the effective opening intensity, shifting the local mode mixity and thereby lowering the observed kink angle. Third, non-singular stresses (the T -stress) and finite process-zone effects can bias the apparent initiation direction: the maximum hoop-stress criterion, based purely on the singular (K_I, K_{II}) fields, neglects the influence of a nonzero T -stress and any inelastic or micro-cracking zone, both of which can rotate the direction of maximum tensile stress and tend to reduce the initial kink angle in confined geometries. Additional contributors include uncertainty in the actual initial flaw geometry (notch root radius, slight asymmetry, or pre-crack curvature), differences between the assumed and realized loading distribution on the disk (finite arc contact and load eccentricity), and experimental measurement bias in defining the “first-segment” angle from a short, possibly curved extension. Taken

together, these effects provide a consistent explanation for why the experimental angles are systematically smaller: real specimens experience contact/friction and confinement that reduce the effective shear-driven component and modify the near-tip stress state relative to the idealized, traction-free LEFM model used in the simulations.

Computational cost and scalability

Wall-clock cost

Because only the crack surface Γ and, in bounded domains, the outer boundary $\partial\Omega$ are discretized, the total number of unknowns in a VCEM simulation is

$$N_{\text{dof}} = N_c N_s + N_{\text{bem}} + m, \quad (30)$$

where N_c is the number of crack elements, $N_s \in \{1, 2, 3\}$ the number of shape-function DOFs per element (piecewise-constant, linear, or quadratic; Sec. 3.1 of our paper (Ghoniem 2026a)), N_{bem} the number of boundary panels, and m the total number of junction-closure and contact multipliers in the KKT system (4)–(5). For all benchmark and finite-domain simulations of [Method validation](#) and [Simulations of disk compression experiments](#) sections, N_{dof} remains of order 10^2 – 10^3 . For comparison, a conventional FEM simulation of the same problem capable of reproducing the $r^{-1/2}$ tip asymptotics requires $N_{\text{FEM}} \sim 10^5$ – 10^6 DOFs, even with singular elements or local enrichment. The dimensional reduction is therefore two to three orders of magnitude at the level of primary unknowns and is preserved despite the addition of equality, inequality, and BEM-coupling constraints, since the constraints contribute only $m \sim 10$ – 10^2 scalar multipliers for the network sizes considered here.

Each crack-growth cycle decomposes into three contributions,

$$T_{\text{cycle}} = T_{\text{asm}} + T_{\text{KKT}} + T_{\text{grow}}, \quad (31)$$

corresponding to kernel assembly, the constrained equilibrium solve, and the growth step (per-tip SIF extraction, kink-angle evaluation, and graph update). In the present implementation, the dense crack–crack and BEM–crack blocks are assembled directly with double quadrature (Galerkin) or single quadrature (least-squares; Sec. 4.4 of our paper (Ghoniem 2026a)), giving $T_{\text{asm}} = \mathcal{O}(N_{\text{dof}}^2)$ in time and storage. The KKT saddle-point system (4) is factorized by direct LAPACK routines, giving $T_{\text{KKT}} = \mathcal{O}(N_{\text{dof}}^3)$. The growth step scales as $T_{\text{grow}} = \mathcal{O}(N_t)$ where N_t is the number of active tips, and is always negligible relative to $T_{\text{asm}} + T_{\text{KKT}}$ for the simulations reported here.

For the problem sizes encountered in this work ($N_{\text{dof}} \lesssim 10^3$), direct dense solvers are entirely adequate. For substantially larger networks, the asymptotic cost can be reduced without altering the formulation: the Kelvin-based kernel is translation-invariant in global coordinates and therefore directly compatible with hierarchical compression schemes (fast multipole methods, $\mathcal{H}$ - and $\mathcal{H}^2$ -matrix representations), reducing T_{asm} and the storage to $\mathcal{O}(N_{\text{dof}} \log N_{\text{dof}})$. The symmetric saddle-point KKT system is amenable to block and null-space preconditioners (Benzi et al. 2005) that exploit the (0, 0) block structure, so that, combined with a fast matrix–vector product, the solver scales as $\mathcal{O}(N_{\text{dof}} \log N_{\text{dof}})$ per Krylov solve. These scalings are the established complexity of modern fast BEM (Liu 2009), and because VCEM shares the BEM kernel structure, they apply directly. A fast-multipole implementation is the subject of ongoing work and is not used in the simulations of this paper.

Table 2 reports the measured wall-clock cost for representative problems in the manuscript, all run single-threaded on a workstation with an Intel Xeon W-2245 CPU (3.9 GHz, 64 GB RAM). Per-cycle times are reported for runs at the discretization actually used to produce the figures in the manuscript ($N_{e,\text{half}} = 40\text{--}90$ panels per half-crack, $N_{\text{bem}} = 120$ Brazilian-disk boundary elements, $N_{q,\text{col}} = 12$, $N_{q,\text{stress}} = 16$, cheb_quad basis with tip_dense clustering, half crack mode, core junction model, iterative crack–BEM coupling).

The most expensive simulation in the manuscript is the disk fragmentation run of [Fragmentation of compressed disks](#) section, in which the crack count and the number of active tips grow continuously through the simulation. At the final stages, with four polyline cracks containing eight active tips, several junctions formed by branching and intersection, and an $N_c \sim 720$ panel budget under iterative BEM coupling, the per-cycle time approaches roughly 25 min on a single core; a full multi-cycle fragmentation sequence therefore completes in several hours of wall-clock time. We emphasize that this cost is dominated by the dense $\mathcal{O}(N_c^2)$ influence-matrix assembly in Python rather than by the KKT solve itself, and that an existing C++/Eigen scaffold (`fracture_utils/Ubem/cpp/`) for the Kelvin kernels and BEM operators is in place but not yet activated for the production runs reported here; preliminary tests indicate a 1–2 order-of-magnitude reduction in cycle time once that path is enabled. The entire manuscript, including all convergence studies and parameter sweeps, was produced on a single workstation.

Memory is dominated by the dense crack–crack and BEM–crack blocks of the KKT matrix, which require $\mathcal{O}(N_{\text{dof}}^2)$ storage — on the order of tens of MB for the simulations reported here and can be reduced to $\mathcal{O}(N_{\text{dof}} \log N_{\text{dof}})$ with hierarchical compression. Graph-based topology updates (edge splitting at intersections, junction creation, boundary-termination handling) are $\mathcal{O}(N_c)$ in the current implementation and have never been a bottleneck in the cases reported.

Comparable simulations of disk fragmentation by alternative methods require substantially larger computational resources for the same near-tip fidelity. Conventional FEM simulations require either volumetric remeshing at every propagation step to keep the mesh aligned with the evolving crack geometry or local refinement around each tip;

Table 2 Representative problem sizes and per-cycle wall-clock times on a single core. N_c is the total number of crack-element panels summed over all polylines in the network at the reported cycle (so $N_c = 180$ corresponds to a single polyline with $N_{e,\text{half}} = 90$, while $N_c \sim 720$ corresponds to four growing polylines with eight active tips and several junctions). N_{bem} is the number of Brazilian-disk boundary elements (“–” when the loading is applied analytically). Cycle time includes panel re-discretization, dense kernel assembly, KKT-constrained least-squares solve, BEM–crack iterative coupling correction (where applicable), Williams-fit SIF extraction at all active tips, propagation-direction selection, and graph-topology update (intersection detection, vertex merge, junction insertion).

Problem	N_c	N_{bem}	N_{dof}	T_{cycle}
Straight mode-I crack (Fig. 2)	100	–	$\sim 10^2$	< 5 s
Inclined biaxial crack (Fig. 3)	120	–	$\sim 10^2$	< 5 s
Symmetric branched crack (Fig. 5)	80	–	$\sim 10^2$	~ 10 s
Asymmetric branched crack (Fig. 4)	240	–	$\sim 10^2$	~ 20 s
Circular arc crack (Fig. 9)	80	–	$\sim 10^2$	~ 10 s
Compressed disk, centerline (Fig. 16)	180	120	$\sim 10^3$	~ 60 s
Compressed disk, off-axis (Fig. 18)	180	120	$\sim 10^3$	~ 90 s
Disk fragmentation (Fig. 23)	~ 720	120	$\sim 3 \times 10^3$	~ 1500 s (~ 25 min)

in either case, the volumetric mesh dominates the DOF count, and the remeshing or refinement cost typically dominates the total wall time as the network evolves (Bouchard et al. 2003). XFEM removes the mesh-alignment requirement but introduces enrichment that must be re-partitioned at every cycle as new junctions appear; published XFEM studies of comparable branched-crack problems report 10^4 – 10^5 DOFs per cycle, with the conditioning of the enriched system deteriorating as the number of interacting cracks grows (Menk and Bordas 2011; Bechet et al. 2005). In phase-field fracture, the diffuse representation requires a regularization length ℓ_0 that must be small compared to the smallest geometric feature of the problem, and the mesh size in the vicinity of every crack must in turn be a fraction of ℓ_0 — typically at most $\ell_0/2$ (Bourdin et al. 2008; Miehe et al. 2010). Because this constraint applies throughout the volume in which any crack may nucleate or propagate, phase-field simulations of multiple interacting cracks require very fine volumetric meshes and incur correspondingly high computational costs, especially in three dimensions (Bourdin et al. 2014; Doitrand et al. 2023). Phase-field approaches must additionally solve a coupled displacement–damage system at every step. Against these baselines, the $\sim 10^3$ degrees of freedom and seconds-per-cycle wall time of VCEM constitute a substantial reduction in computational cost while retaining the sharp-crack near-tip asymptotics analytically.

User-controlled and adaptive crack-element refinement

The current VCEM implementation already provides several non-adaptive discretization options that allow the user to control the number of degrees of freedom along each crack: uniform spacing, tip-dense (clustered) spacing, Chebyshev–Lobatto clustered nodes (Sec. 3.1 of our companion paper (Ghoniem 2026a)), and the singular tip representation that embeds the $1/\sqrt{r}$ asymptotics directly into the basis functions. These options control DOF placement *a priori* — they concentrate elements near tips and junctions where the fields are most singular — but they do not respond automatically to the computed solution. Tip-density and clustering parameters are set at the start of a simulation and held fixed throughout the growth sequence.

A genuine *adaptive* extension, in the sense of the global–local mesh-density mapping of Cheng et al. (2025), would close the loop by driving local refinement from a-posteriori error indicators computed from the converged solution at each cycle. In the VCEM setting, the natural error indicator is the residual of the Williams-fit SIF extraction (Sec. 3.1.1) over the near-tip window: tips at which the Williams residual exceeds a prescribed tolerance would trigger local h-refinement of the adjacent crack elements, while tips well-resolved by the current discretization would be left unchanged. This is the boundary-integral analog of the global–local mesh-density mapping proposed by Cheng et al., and it is naturally compatible with the graph-based topology updates already implemented (Sec. 6 of our companion paper (Ghoniem 2026a)) — subdividing an edge in the crack graph is the same operation already used to insert junctions at intersection points. The adaptive extension is identified here as a natural follow-up to the user-controlled discretization options already in the code, but it is not exercised in the simulations of the present paper.

Outer-cycle convergence and termination

The iterative BEM–KKT scheme is driven by an outer loop that alternates between (i) a constrained crack solve through the direct KKT coupling, advancing the network by up to $N_{\text{in}} = 6$ growth increments per outer pass, and (ii) a boundary-equilibrium update in which the crack-induced tractions on the disk boundary, $\mathbf{t}_{\text{cr}} = \boldsymbol{\sigma}_{\text{cr}} \cdot \mathbf{n}$, are evaluated at the BEM collocation points from the current Burgers density and re-applied as reversed corrections $-\mathbf{t}_{\text{cr}}$ to the BEM solve. The full update is used at each outer cycle (no under-relaxation), since the BEM and KKT subproblems are linear and the crack-induced perturbation on the boundary is small relative to the applied diametral load.

Convergence is monitored a posteriori through the relative change of the boundary-traction correction between successive outer cycles,

$$r^{(k)} := \frac{\|\mathbf{t}_{\text{cr}}^{(k)} - \mathbf{t}_{\text{cr}}^{(k-1)}\|_2}{\|\mathbf{t}_{\text{cr}}^{(k-1)}\|_2 + \varepsilon_0}, \quad \varepsilon_0 = 10^{-30}, \text{ Pa},$$

and through the corresponding relative change of the disk boundary compliance $C_{\text{rel}}^{(k)}$. For all simulations reported in [Centerline slit: growth, closure, and shear localization–Compressed disk with an off-axis slit](#) sections we used a fixed budget of $N_{\text{out}} = 8$ outer cycles, which was sufficient to drive $r^{(k)}$ below 10^{-3} and the relative change in C_{rel} below 10^{-4} before the final outer cycle. The sequence $r^{(k)}$ was found to decrease monotonically in every reported run; no oscillatory or divergent behavior was observed. With these settings, a complete simulation of [Centerline slit: growth, closure, and shear localization–Compressed disk with an off-axis slit](#) sections requires $N_{\text{out}} \times N_{\text{in}} = 48$ growth increments and the same number of KKT solves, plus $N_{\text{out}} = 8$ BEM updates.

Fracture as a “connectivity phase transition”

Order parameters

Fracture can be interpreted within an energy-based framework that closely parallels the structure of thermodynamic phase transitions. When fracture proceeds through a distributed population of interacting cracks, a different description becomes appropriate. Instead of a single geometric variable, the system is characterized by collective measures of damage and connectivity. We represent a crack network in a bounded domain $\Omega \subset \mathbb{R}^2$ (e.g., a disk) as a graph $G = (V, E) = \bigcup_{e=1}^{N_c} \Gamma_e$, where V denotes the set of vertices and E the set of crack edges, each Γ_e being a parametric curve segment of length ℓ_e . The total crack length is $L_{\text{tot}} = \sum_{e \in E} \ell_e$. The graph decomposes into N_c connected components $G = \bigcup_{k=1}^{N_c} G_k$, each G_k being a maximal connected subgraph. The length of component G_k is $L_k = \sum_{e \in G_k} \ell_e$. Let $B = \partial\Omega$ denote the external boundary. A connected component G_k is said to be *spanning* if it intersects at least two disjoint boundary regions,

$$G_k \cap B_1 \neq \emptyset, \quad G_k \cap B_2 \neq \emptyset, \quad B_1 \cap B_2 = \emptyset. \quad (32)$$

In a diametrically compressed disk, for example, a spanning cluster connects the upper and lower loading arcs, thereby forming a system-scale fracture path.

Let G_{span} denote the spanning component (if it exists), with total length $L_{\text{span}} = \sum_{e \in G_{\text{span}}} \ell_e$. We define the spanning cluster fraction

$$P_{\infty} = \frac{L_{\text{span}}}{L_{\text{tot}}}. \quad (33)$$

If no spanning cluster exists, $P_\infty = 0$.

The following order parameters are introduced:

1. Crack surface density (damage variable):

$$\alpha = \frac{\sum_i \ell_i}{A}$$

where ℓ_i are crack segment lengths.

2. Spanning cluster fraction:

$$P_\infty = \frac{\text{crack length in spanning clusters}}{\text{total network crack length}}$$

The quantity P_∞ measures global crack connectivity. For $P_\infty = 0$, cracks remain isolated and damage is distributed. When $P_\infty > 0$, a macroscopic fracture path has formed, producing strong compliance growth and stress localization. In analogy with percolation theory, P_∞ serves as a connectivity order parameter, while the total crack density α measures the magnitude of damage. The emergence of P_∞ signals the transition from distributed microcracking to system-spanning fracture.

3. Alignment parameter:

$$Q = \langle 2 \cos^2 \theta - 1 \rangle$$

where θ is the angle between crack segments and the loading axis. $Q = 1$ corresponds to perfect alignment, and $Q = 0$ to isotropic orientation.

To illustrate the evolution of crack network damage metrics, we go back to the case of the inclined slit of [Compressed disk with an off-axis slit](#) section, but we will change the loading condition to “boundary displacement” control. The top arc is displaced $8 \mu\text{m}$ relative to the bottom arc in the vertical direction, while all other conditions are the same as in [Compressed disk with an off-axis slit](#) section. In this case, we expect that once the crack propagates, the compressed disk becomes more *compliant*, and boundary forces are relaxed. These conditions may be conducive to the arrest of a growing crack after it has already started to grow. Under force control, this may be possible only under very idealized circumstances, but with displacement control, crack arrest may be more easily achieved.

Figure 24 shows the evolution of crack-network damage metrics during the propagation of an initially inclined crack in the compressed disk under boundary-displacement control. The cumulative crack length (or crack density α) in panel (a) increases rapidly during the early growth cycles and then approaches a plateau as crack propagation slows. Panel (b) shows the evolution of the alignment parameter $Q = \langle 2 \cos^2 \theta - 1 \rangle$, which increases as the crack trajectory rotates toward the preferred tensile splitting direction and then stabilizes once the crack geometry becomes stationary. Panel (c) shows the corresponding evolution of the crack-tip stress intensity factor K_{tip} relative to the fracture toughness K_{Ic} . Initially $K_{\text{tip}} > K_{Ic}$ driving crack extension; however, as the crack grows, the compliance of the disk increases, reducing the load required to maintain the

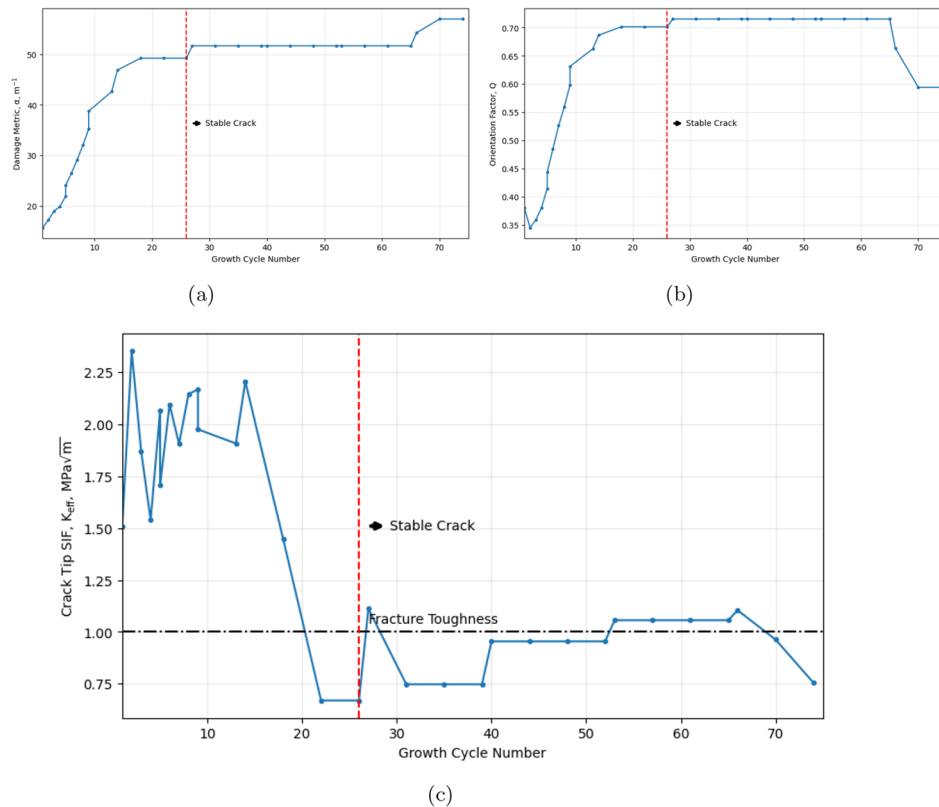


Fig. 24 Evolution of network damage metrics. Note the crack growth has led to increased disk compliance, as manifest in the reduction of the tip SIF below the material's fracture toughness, leading to crack arrest

imposed displacement. As a result, K_{tip} decreases and eventually falls below K_{Ic} , leading to crack arrest.

Fragmentation of compressed disks

At low crack density α , cracks relax energy locally and remain isolated. As α increases, long-range elastic interactions promote coalescence. At a critical value α_c , a spanning cluster forms, $P_\infty > 0$, and the system undergoes macroscopic localization. This resembles a percolation transition in which connectivity serves as the order parameter.

Two limiting behaviors are possible:

- *Continuous (percolation-like) transition:* P_∞ grows gradually from zero, and cluster statistics exhibit scale-free behavior. Energy release accelerates but does not jump discontinuously in the infinite-system limit.
- *Discontinuous (localization) transition:* interactions drive rapid coalescence into a dominant crack, producing a sharp drop in Φ and a sudden increase in Q . This resembles a first-order transition with avalanche-like energy release.

To connect the phase-transition viewpoint to the compressed disk, it is useful to write a simplified—but mechanically faithful—expression for the energetic functional in terms of the crack density α . For a linear elastic body subjected to a scalar load parameter P with a corresponding generalized displacement δ , define the compliance, $C(\alpha) = \frac{\delta}{P}$. Since crack growth generally increases structural flexibility, $C(\alpha)$ is an increasing function of α , i.e. $dC/d\alpha > 0$. Under traction or force control (e.g. prescribed pressure on

loading arcs), the resultant load P is fixed while the displacement δ adapts. The stored elastic energy is

$$U(\alpha) = \frac{1}{2}P\delta = \frac{1}{2}P^2C(\alpha). \quad (34)$$

For dead loads, the external work is $W_{\text{ext}}(\alpha) = P\delta = P^2C(\alpha)$, so the total potential energy becomes

$$\Phi_{\text{force}}(\alpha) = U(\alpha) - W_{\text{ext}}(\alpha) = -\frac{1}{2}P^2C(\alpha). \quad (35)$$

Because $dC/da > 0$, it follows that $\Phi_{\text{force}}(\alpha)$ decreases with crack extension. The corresponding energy release rate is

$$G_{\text{force}}(\alpha) = -\frac{d\Phi_{\text{force}}}{dA} = \frac{1}{2}P^2\frac{dC}{dA}. \quad (36)$$

Thus, under force control, increasing compliance directly lowers the potential energy, which can promote unstable crack growth.

Under boundary displacement control, however, the generalized displacement δ is prescribed while the load P adapts accordingly, $P(\alpha) = \frac{\delta}{C(\alpha)}$. The stored elastic energy is now

$$U(\alpha) = \frac{1}{2}P\delta = \frac{1}{2}\frac{\delta^2}{C(\alpha)}. \quad (37)$$

In displacement control, the external work is not an independent variable; the appropriate energetic functional is simply the stored energy.

$$\Phi_{\text{disp}}(\alpha) = U(\alpha) = \frac{1}{2}\frac{\delta^2}{C(\alpha)}. \quad (38)$$

Since $C(\alpha)$ increases with crack length, $\Phi_{\text{disp}}(\alpha)$ decreases more gradually, and its curvature fundamentally differs from the force-controlled case. The corresponding energy release rate is

$$G_{\text{disp}}(\alpha) = -\frac{d\Phi_{\text{disp}}}{dA} = \frac{1}{2}\frac{\delta^2}{C(\alpha)^2}\frac{dC}{dA}. \quad (39)$$

The two expressions differ based on the ensemble in which the system is viewed:

$$G_{\text{force}}(\alpha) = \frac{1}{2}P^2\frac{dC}{dA}, \quad (40)$$

$$G_{\text{disp}}(\alpha) = \frac{1}{2}\frac{\delta^2}{C^2}\frac{dC}{dA}. \quad (41)$$

Under force control, crack extension always reduces the total potential energy, which can lead to snap-through or discontinuous instability. Under displacement control, the energetic decrease is moderated by the inverse dependence on compliance, often resulting in more stable, gradual crack growth. These differences reflect the distinct Legendre structures associated with force and displacement control and play a central role in interpreting fracture as a phase-transition-like phenomenon. Crack growth incurs a

surface energy cost. If $A(\alpha)$ denotes the total crack surface (or length in 2D), the relevant thermodynamic potential is

$$\mathcal{F}(\alpha) = \Phi(\alpha) + \Gamma A(\alpha), \quad (42)$$

where Γ is the fracture energy. The evolution condition becomes

$$\frac{d\mathcal{F}}{d\alpha} = 0, \quad \text{with} \quad G(\alpha) = -\frac{d\Phi}{dA} \geq \Gamma. \quad (43)$$

To understand the complexity of crack network evolution, let's consider the case of an initial distribution of cracks in the same disk outlined in [Compressed disk with an off-axis slit](#) section, where we can either apply a dead-load of 3.84 kN at the top and bottom arcs (force control). When a dead-load is applied as a *tuning* parameter and its magnitude is above a critical value, some or all crack tips will start growing, depending on the local energy landscape surrounding each one. Once various branches grow, the effective SIF at each tip cannot be reduced because the load does not relax, and longer cracks acquire higher effective SIFs. One way that crack tips may be arrested is if a tip can get trapped in a locally shielded region of the disk; otherwise, the crack network system is energetically *unstable*, and most or all branches will eventually reach the disk boundaries, leading to disk “fragmentation”.

Before presenting the fragmentation simulations, we note that the crack-growth algorithm used throughout this section is the same per-tip rule developed in [Crack growth algorithm](#) section, applied without modification to every active tip of the evolving network. At each cycle, the constrained equilibrium problem is solved for the full network, and the near-tip SIFs ($K_I^{(i)}$, $K_{II}^{(i)}$) are extracted at every active tip $\tau_i \in \mathcal{T}^{(n)}$ from the converged interacting fields. The per-tip kink angle θ_i and the per-tip increment Δa_i are then computed from Eqs. (27) and (29) respectively, and all active tips are advanced simultaneously within the same cycle. Crack-crack interactions — shielding, amplification, and junction-mediated load redistribution — are therefore resolved through the global equilibrium solve rather than through any ordering of the growth increments. The number of simultaneously active tips grows from two in the initial configuration to more than ten in the final stages of Figs. 25 and 26, and all are advanced in parallel at every cycle. Intersections detected during the geometric update insert new junctions and trigger re-assembly of the junction-closure constraints $C_{Jq} = 0$, so that the Burgers-vector conservation condition is restored at the start of the next cycle. This is what allows the simulation to transition smoothly from a collection of isolated cracks ($P_\infty = 0$) to a system-spanning fracture network ($P_\infty = 1$) without special handling of branching or coalescence events: the equilibrium solve, the per-tip rule, and the topology update are applied at every cycle regardless of how simple or complex the current configuration is.

Figures 25 and 26 illustrate the progressive fragmentation of the compressed disk when the system is subjected to dead-load (force) control. In this loading mode, the external force remains fixed, so any increase in structural compliance caused by crack extension does not reduce the applied driving force. Consequently, once the load exceeds a critical value, crack growth becomes energetically unstable and propagates through the network until the disk is fragmented. Figure 26 shows the geometric evolution of the crack network. Panels (a)–(c) present the graph representation of the crack network at three stages of growth: the initial defect configuration, an intermediate stage where

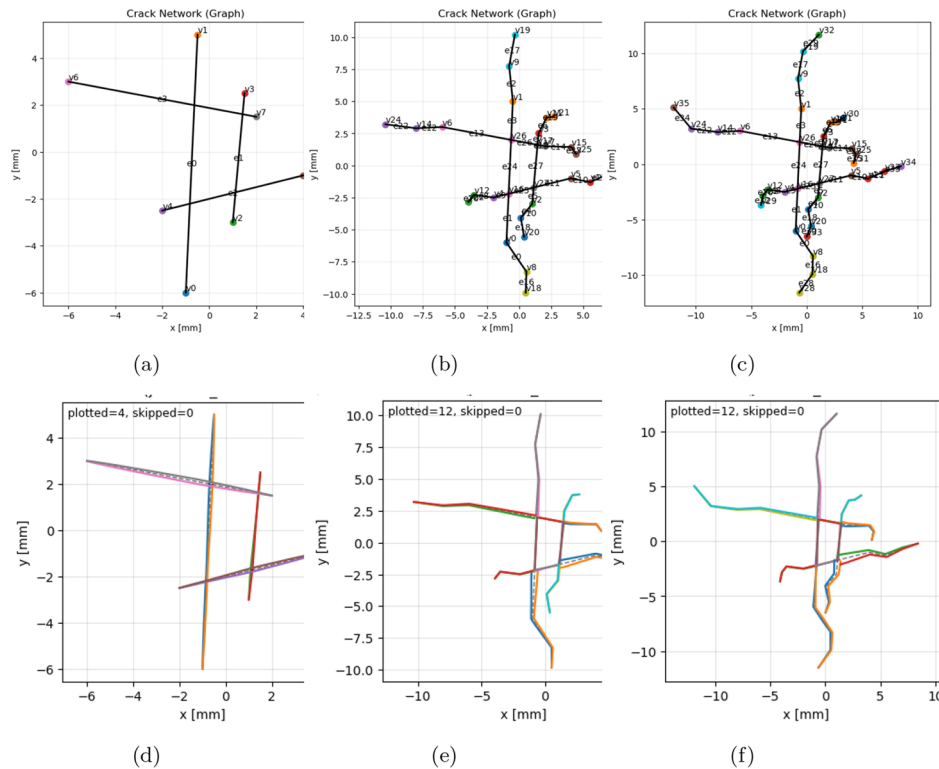


Fig. 25 Crack network evolution of a compressed disk: **a** Initial defect (crack) configuration represented as a graph, **b** intermediate network graph structure, **c** final network graph structure. Note the detection of intersections and the distributions of vertices of degrees 1 (tips), 2 (kinks), and 3 (junctions). **d** initial deformed crack network configuration constructed from the COD/CSD profiles. **e** intermediate deformed configuration. **f** final crack network deformed configuration

several branches begin to extend and interact, and the final configuration in which multiple cracks intersect and form a connected network. During this process, intersections are detected, and the topology evolves through the appearance of vertices of different degrees corresponding to crack tips (degree 1), kinks (degree 2), and junctions (degree 3). Panels (d)–(f) show the corresponding deformed crack configurations obtained from the computed crack-opening and crack-sliding displacement fields. The progressive extension of branches and the formation of junctions lead to a network that spans the disk and divides it into several fragments. The transition from isolated cracks to a connected fracture network reflects the cooperative nature of crack interactions in the disk.

Figure 26 shows the corresponding evolution of the stress field during the fragmentation process. Panels (a) and (c) present the stress components σ_{xx} and σ_{xy} during the early stages of loading, when cracks are still isolated and the spanning cluster fraction P_∞ is zero. In this regime, the stress field is distributed over the disk, and only localized perturbations appear near individual crack tips. As the network evolves, crack growth redistributes the stress field and strongly localizes stresses near the advancing tips. Panels (b) and (d) show the stress field after the network has grown to form a connected fracture path spanning the disk. At this stage, the spanning cluster fraction becomes $P_\infty = 1$, indicating that a system-scale fracture path has formed. The stress field becomes largely relaxed in the bulk of the disk while remaining concentrated along the newly formed crack network.

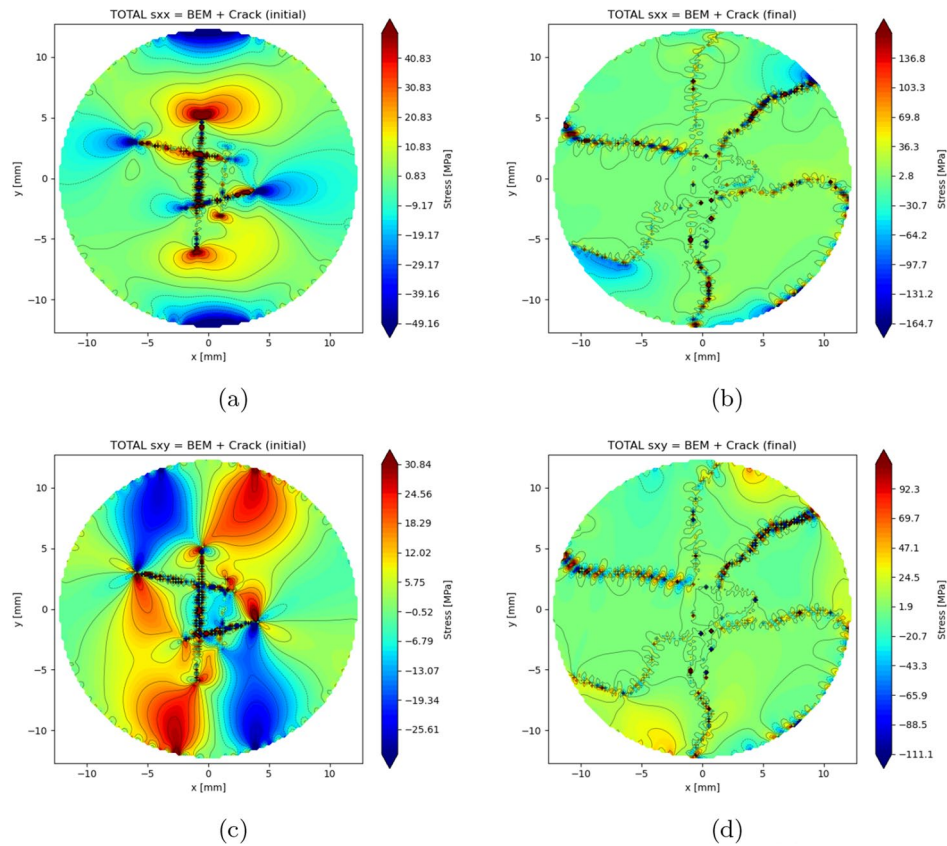


Fig. 26 Evolution of the stress field in the compressed disk as it undergoes multiple fractures: **a** σ_{xx} at the initial stages of disk compression. **b** σ_{xx} at the final stages of disk fracture. **c** σ_{xy} at the initial stages of disk compression. **d** σ_{xy} at the final stages of disk fracture. Note the stress redistribution in the disk, being relaxed everywhere, but getting concentrated close to crack tips. Note the transition from $P_\infty = 0$ in 26a and 26c to $P_\infty = 1$ in 26b and 26d

The transition from $P_\infty = 0$ to $P_\infty = 1$ represents the emergence of a system-spanning fracture cluster, analogous to the appearance of a percolating cluster in statistical physics. Under force control, the absence of load relaxation allows crack growth to proceed until this connectivity transition is reached, resulting in catastrophic fragmentation of the disk. In this sense, fragmentation can be interpreted as a phase-transition-like instability driven by the external load acting as a control parameter and the crack connectivity serving as the order parameter.

Conclusions

This paper has presented a systematic validation of the Variational Crack Element Method (VCEM) introduced in our companion paper (Ghoniem 2026a), in which cracks are represented as distributed Burgers density fields embedded within an energy-minimization structure that preserves the singular near-tip asymptotics while enabling the treatment of interacting crack geometries within a unified computational framework.

Validation was performed at three levels of geometric complexity. For straight cracks under pure mode-I, mode-II, and mixed-mode biaxial loading, numerical stress intensity factors and crack-tip displacement fields agree closely with classical analytical solutions, confirming that the distributed Burgers density representation correctly captures the singular structure of the near-tip fields and the coupling between fracture modes. For

kinked, symmetrically branched, and circular-arc cracks, the method reproduces published analytical results derived from complex-variable formulations and the theory of Cotterell and Rice, demonstrating accuracy under abrupt changes in crack orientation and curvature. Across all of these benchmarks, the formulation remains stable under alternative parametric representations—polyline segments, circular arcs, and cubic splines—which are unified through a graph-based topological description that treats tips, kinks, and junctions consistently.

The method was then applied to the diametrical compression of a disk, a finite-domain problem with strongly non-uniform stress fields under both force and displacement control. For a centerline crack, VCEM reproduces the classical splitting behavior, with the propagating tips arrested as they enter the compressive zones beneath the loading arcs. For inclined cracks, the simulated trajectory rapidly reorients toward the maximum-opening direction, and the predicted kink angles agree quantitatively with experimental measurements. Finally, for configurations containing multiple interacting defects, the simulations reveal that crack-induced compliance growth under force control produces unstable propagation, junction formation, and ultimately fragmentation of the disk into several distinct regions. This collective evolution admits a natural interpretation in terms of connectivity transitions: as cracks coalesce into a domain-spanning cluster, the system undergoes a transition analogous to percolation, with the applied load as the control parameter and the connectivity of the crack network as the order parameter.

Two features of the formulation deserve emphasis in this closing discussion, as they underpin the broader applicability of the method. The first is the absence of an intrinsic length scale: the Kelvin-based kernel depends only on the dimensionless geometric quantity Δ/R and on Poisson's ratio, with the shear modulus entering as an overall amplitude. The same code base therefore applies from the sub-micron to the meter scale without modification, and the multiscale structure of crack fields — $r^{-1/2}$ near the tip, dipolar at intermediate distances, and multipolar in the far field — is reproduced automatically through the kernel without volumetric refinement. The straight-crack convergence study demonstrates this explicitly, recovering K_I to within 1% over crack lengths spanning four orders of magnitude using a single discretization strategy.

The second feature is the equilibrium–growth separation built into the per-tip propagation rule. Because crack interactions are resolved through the global KKT solve while crack advance is determined locally at each tip from the converged fields, any microstructural information that can be expressed as a spatially varying field can be incorporated through the local rule without modifying the equilibrium solver. The formulation already supports a spatially varying fracture toughness $K_c(\mathbf{x})$ for representing grain-boundary toughness, inclusions, or laminate architectures; stochastic propagation thresholds obtained by sampling K_c from a prescribed random field; and heterogeneous local criteria, in which each tip may use a different fracture rule (MTS, local-symmetry, or SED) according to its microstructural context. The validation simulations of the present paper use homogeneous elastic constants and toughness and therefore do not themselves exercise this capability; a systematic study of microstructural effects, including coupling to lattice dislocation fields and orientation-dependent toughness derived from local texture descriptors, is the subject of ongoing work.

Beyond the purely mechanical loading considered here, several coupled fracture and interface-failure problems provide natural application classes for the formulation. A

particularly direct example is hydraulically driven fracture in geomaterials, where recent laboratory and theoretical studies have shown that fluid-driven cracks under anisotropic confinement can undergo unstable curving, branching, and network-like growth patterns resembling the multi-crack fragmentation regime (Li et al. 2025; Xu et al. 2026). Closely related mechanisms also arise in ion-implantation-induced blistering, thin-layer splitting, and coating-delamination problems, where subsurface gas accumulation, implantation damage, residual stress, and interface adhesion jointly drive crack nucleation and propagation parallel or oblique to material interfaces (Singh et al. 2010; Galindo et al. 2005). These processes are important in compound semiconductors, thin films, plasma-facing materials, protective coatings, and layered functional materials, where fracture is governed not only by the external loading but also by internal pressure, interface toughness, adhesion strength, and evolving defect or gas concentrations.

The connection between VCEM and this broader class of problems is direct. The essential solid-mechanics building blocks—distributed crack representation, network topology with kinks and junctions, unilateral contact, interface constraints, and criterion-based propagation—are already present in the formulation. For hydraulic fracture, the crack-face loading can be generalized by replacing the prescribed mechanical traction with a time-dependent fluid pressure $p_f(\mathbf{x}, t)$ obtained, for example, from a one-dimensional Reynolds-type flow equation along each crack together with a leak-off model into the surrounding matrix. For blistering or adhesion-driven delamination, the same framework can be extended by prescribing an internal gas pressure, swelling eigenstrain, or implantation-induced stress field, together with an interface fracture criterion involving the local adhesion energy or mixed-mode interfacial toughness. In each case, the right-hand side of the KKT system becomes time-dependent through the applied pressure, eigenstress, or interface traction, while the junction-closure constraints $C_{Jq} = 0$ continue to enforce displacement single-valuedness and Burgers-vector closure as cracks curve, branch, coalesce, or delaminate. Coupling VCEM to fluid-flow, poro-elastic, gas-transport, or adhesion-interface models along these lines is therefore a natural extension of the present work and is left to follow-up studies.

The present formulation is restricted to two dimensions by design in order to isolate the constrained-variational structure — KKT junction closure, unilateral contact, BEM coupling, and graph-based topology updates — in a setting where the crack representation, interaction kernels, near-tip extraction, and topological updates can all be developed without the additional complications of three-dimensional crack-front geometry. The 2D formulation is nonetheless directly relevant to thin-plate fracture, plane-strain idealizations of laminate or coating structures, and 2D cross-sections of rock specimens used in disk-compression and similar tests. Several features of the present formulation generalize naturally to three dimensions: the Kelvin fundamental solution has a direct 3D analog, the constrained energy-minimization principle is dimension-independent, and the equilibrium–growth separation together with the graph-based topology extends to a 2-complex in which crack tips become crack *fronts* (1D curves on a crack surface) and junctions become intersection lines of multiple surfaces. Other features require additional formulation work: the Burgers density becomes a vector-valued surface field rather than a line field, the Williams expansion at the front carries an additional out-of-plane mode (mode III), so the SIF extraction must distinguish three modes rather than two, and the propagation kinematics involve front-velocity fields rather than per-tip

increments. Computationally, the dense kernel blocks scale as $\mathcal{O}(N_{\text{dof}}^2)$ as in 2D, but N_{dof} grows with crack *area* rather than length, raising problem sizes by roughly two orders of magnitude for comparable geometric resolution; this is the regime where the hierarchical compression schemes discussed in [Computational cost and scalability](#) section (fast multipole, $\mathcal{H}$ -matrix) become essential rather than optional.

Taken together, these properties make VCEM structurally compatible with micro-structure-informed fracture simulation across a wide range of length scales, as well as with the extensions to fluid-driven and three-dimensional fracture identified above. Demonstrating these capabilities for specific material systems — composite structures, additively manufactured materials, microelectronic devices, and plasma-facing components in fusion energy systems — is a natural direction for subsequent work, and the structure that would enable such extensions is built into the formulation itself.

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Authors' contributions

Nasr Ghoniem is the sole author of this manuscript. He conceptualized the key ideas, formulated the solutions, performed computer simulations, obtained all results, wrote the paper, and reviewed all details.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

The author declares that this work is original, has been conducted in accordance with accepted standards of research integrity, and does not involve human or animal subjects.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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